A Unified Framework for Dynamic Čech Cohomology and Sheaf Theory with Applications in TDA

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Abstract

This paper addresses the theoretical limitations of traditional topological data analysis (TDA) in handling dynamic systems, and constructs a unified framework for dynamic Čech cohomology and sheaf theory on manifolds. By modeling dynamic processes as continuous simplicial flows, we define dynamic Čech complexes and dynamic Čech sheaves, and establish a mathematical description for the continuous evolution of topological features. We innovatively propose the theory of persistent dynamic cohomology, and introduce dynamic topological feature measures to circumvent the indecomposability problem of multi-parameter persistence modules. An efficient incremental computation algorithm is developed to realize the topological feature analysis in the scale-time dual-parameter space. Experiments on human motion capture data demonstrate that the proposed framework can effectively capture the dynamic topological features of different motion patterns, and the Wasserstein distance matrix verifies its potential in pattern recognition. This work provides a new theoretical foundation and computational method for dynamic topological data analysis, and has promising applications in fields such as biological motion analysis and complex system monitoring.

Keywords: Dynamic Topological Data Analysis, Čech Cohomology, Sheaf Theory, Persistent Homology, Manifold Learning, Motion Analysis

Contents

1 Introduction **1**

2 Preliminaries **3**
- 2.1 Complexes and Persistent Homology 3
- 2.2 Čech Cohomology and Sheaves 7

3 Dynamic Čech Complexes and a Sheaf-Theoretic Unified Framework **9**
- 3.1 Theory of Dynamic Čech Complexes 9
- 3.2 Construction of the Dynamic Čech Sheaf 19

4 Persistent Dynamic Cohomology Theory 25
4.1 Fundamental Theory of Persistent Dynamic Cohomology 26
4.2 Convergence, Stability, and Dynamic Topological Feature Measures 29
4.3 Statistics and Analysis of Dynamic Topological Features 32

5 Dynamic Topological Feature Extraction Algorithms 34
5.1 Efficient Construction Algorithm for Dynamic Čech Complexes . . . 34
5.2 Multi-Parameter Computation Method for Persistent Dynamic Cohomology 36
5.3 Measure-Based Topological Feature Extraction and Visualization Algorithms 39

6 Conclusion and Outlook 40

A Proof of Stability Theorem 4.9 and 4.13 42

B Proof of Convergence Theorem 4.10 and its Corollaries 45
B.1 Convergence in the Dynamic Interpolation Distance 46
B.2 Other Convergence Results 46

1 Introduction

In modern data science, the emergence of high-dimensional data poses severe challenges to traditional analytical methods. Although data generated from fields such as sensor detection, gene sequencing, and image feature extraction is presented in high-dimensional form, its intrinsic structure is often constrained by underlying geometric rules—namely, the generated high-dimensional observational data is typically embedded in a low-dimensional manifold structure[CG09]. Topological Data Analysis (TDA) has emerged as a result, and its core problem can be formulated as follows: given a finite point set $X_n = \{x_1, \ldots, x_n\}$ in a high-dimensional observational space $\mathbb{R}^D$, it is essentially a random sample of an unknown d-dimensional compact Riemannian manifold (M, g) under a certain probability distribution μ, where $d \ll D$, i.e., $X_n \overset{\text{i.i.d.}}{\sim} \mu$.

The core task of TDA is to construct a mapping $\mathscr{C}$ from the power set $2^{\mathbb{R}^D}$ to a certain discrete label set $\mathscr{Y}$, such that for a sample X' from an unknown manifold M' and the true label y of M', we have

$$\mathbb{P}\big(\mathscr{C}(X') \neq y\big) < \varepsilon, \quad \varepsilon > 0.$$

Moreover, this mapping $\mathscr{C}$ can preserve the topological invariants of M, thus being robust to noise and redundancy.

To recover the topology of M from discrete samples X', Persistent Homology, systematically proposed by Edelsbrunner et al.[ELZ02], has become a core tool. Its approach involves constructing a Filtered Simplicial Complex and studying the evolution of its homology groups. This method has been successfully applied to numerous fields, ranging from molecular structure analysis to cosmology research[Was16].

Traditional TDA mostly relies on the Vietoris-Rips complex, which is widely used due to its high computational efficiency[Ott+17]. However, the Čech complex provides a more accurate topological approximation through its cover-based geometric construction. Carlsson[Car14] pointed out that the direct connection between the Čech complex and the Nerve Lemma gives it significant theoretical advantages. In recent years, with the improvement of computational

capabilities, the Čech method has regained attention, especially in applications requiring high-precision topological features.

As a powerful tool for describing local-global relationships, Sheaf Theory has been introduced into TDA in recent years. Robinson[Rob14] systematically elaborated on the applications of Sheaf Theory in signal processing in *Topological Signal Processing*. More relevantly, the work of de Silva et al.[SG07] explored the sheaf-theoretic structure of multi-parameter persistent homology, laying the foundation for the in-depth application of Sheaf Theory in TDA.

Topological analysis of dynamic data is a current research hotspot. Existing methods for handling dynamic data are mostly based on discretization and static approximation. Let $I = [0, T]$ be the time parameter interval, and the dataset be $\{X_n(t)\}_{t\in I}$. The mainstream approach is to construct discrete time slices $\{t_0, t_1, \ldots, t_m\}$, establish a static complex for each t_k, and then analyze its persistent homology[PH13]. Although multi-parameter persistent homology provides a more general theoretical framework, it suffers from high computational complexity and faces the fundamental challenge of indecomposability, while failing to fully utilize the geometric properties of the Čech complex.

Despite progress in research, there remain the following essential deficiencies at the mathematical level:

1. **Lack of a continuous dynamic theory**: Existing methods fail to define an intrinsic geometric structure of complexes that continuously depends on parameters. Their analytical objects are discrete sequences $\{\mathcal{K}_{t_k}\}_{k=1}^m$, rather than a continuous "simplicial flow" $\{\mathscr{C}(t)\}_{t\in I}$. This results in the inability to describe the continuous evolutionary paths of topological features and to establish a corresponding differential theory.

2. **Separation of Čech theory and Sheaf Theory**: The strong geometric expressiveness of the Čech complex and the excellent ability of Sheaf Theory to describe local-global relationships have not been unified within a dynamic framework. Specifically, there is a lack of a sheaf-theoretic structure to unify the intuitively "dynamic Čech complex", making it impossible to establish a local-global correspondence principle for dynamic topological features.

3. **Challenge of indecomposability for multi-parameter persistence modules**: Bi-parameter persistence modules are indecomposable in general cases[BS25], and traditional barcode decomposition methods fail, posing a fundamental theoretical obstacle to dynamic topological analysis.

To address the above issues, this paper aims to construct a unified framework for dynamic Čech cohomology and Sheaf Theory on manifolds. The core lies in the mathematical construction at the following three levels:

1. **Dynamic Čech Complex**: For the first time, a rigorous definition of the parameterized Čech complex is given. Considering a continuous parameter space T, a family of Čech complexes $\mathscr{C}(t)$ varying with $t \in T$ is defined, thereby forming a complex-valued functor $\mathscr{C} : T \to \mathsf{Simp}$, and its continuity and differentiability theories are established.

2. **Dynamic Čech Sheaf**: A dynamic Čech sheaf $\mathscr{D}_{\Phi,\varepsilon}$ is constructed on the product space $M \times T$. Through dynamic star neighborhoods and local consistency conditions, a local-global correspondence for the continuous evolution of topological features is established.

3. **Persistent Dynamic Cohomology**: The Persistent Dynamic Cohomology module $\mathbb{PDH}^p(\Phi)$ is defined. By introducing the dynamic topological feature measure μ_Φ^p to characterize the

distribution of topological features in the scale-time parameter space, the indecomposability problem of multi-parameter persistence modules is circumvented, and a stability theorem under the dynamic interpolation distance is proven.

2 Preliminaries

2.1 Complexes and Persistent Homology

Definition 2.1 (Abstract Simplicial Complex). An abstract simplicial complex K consists of a finite set V (called the vertex set) together with a family Σ of subsets of V (called the simplex set) satisfying:

1. For any $v \in V$, we have $\{v\} \in \Sigma$.
2. If $\sigma \in \Sigma$ and $\tau \subseteq \sigma$, then $\tau \in \Sigma$.

Definition 2.2 (p-Simplex). Let K be a simplicial complex. The dimension of a simplex $\sigma \in \Sigma$ is defined as $\dim \sigma = |\sigma| - 1$. A p-simplex of K is a simplex in K of dimension p, i.e., a $\sigma \in \Sigma$ such that $|\sigma| = p + 1$. In particular, 0-simplices are called vertices, 1-simplices are called edges, 2-simplices are called triangular faces, and 3-simplices are called tetrahedra.

Definition 2.3 (Vietoris–Rips Complex). Given a metric space (X, d) and a radius $\epsilon > 0$, the Vietoris–Rips complex of a point set $X_n \subset X$ is defined as:

$$\mathcal{VR}_\varepsilon(X_n) := \{\sigma \subseteq X_n | \forall x, y \in \sigma,\ d(x, y) \leqslant \varepsilon\}.$$

Definition 2.4 (Čech Complex). Given a metric space (X, d) and a radius $\epsilon > 0$, the Čech complex of a point set $X_n \subset X$ is defined as:

$$\mathscr{C}_\varepsilon(X_n) := \left\{ \sigma \subseteq X_n \middle| \bigcap_{x \in \sigma} B(x, \varepsilon) \neq \varnothing \right\}.$$

where $B(x, \varepsilon)$ denotes the open ball centered at x with radius ϵ.

Proposition 2.5 (Inclusion Relations Between Complexes)**.** *For a point set X_n in the same metric space and any $\varepsilon > 0$, we have:*

$$\mathscr{C}_\varepsilon(X_n) \subseteq \mathcal{VR}_{2\varepsilon}(X_n) \subseteq \mathscr{C}_{2\varepsilon}(X_n).$$

Definition 2.6 (Chain Complex). A chain complex $(C_\bullet, \partial_\bullet)$ is a sequence of abelian groups (or modules) C_p ($p \in \mathbb{Z}$) together with group homomorphisms (called boundary operators) $\partial_p \colon C_p \to C_{p-1}$:

$$\cdots \longrightarrow C_{p+1} \xrightarrow{\partial_{p+1}} C_p \xrightarrow{\partial_p} C_{p-1} \longrightarrow \cdots$$

such that $\partial_p \circ \partial_{p+1} = 0$ for all p.

Definition 2.7 (Homology Group). The p-th homology group of a chain complex $(C_\bullet, \partial_\bullet)$ is defined as:

$$H_p(C_\bullet) := {}^{\ker \partial_p}\!/_{\partial_{p+1} C_{p+1}}$$

An element $c \in \ker \partial_p$ is called a p-cycle, and an element $c \in \operatorname{im} \partial_{p+1}$ is called a p-boundary.

Definition 2.8 (Simplicial Chain Complex). Given a simplicial complex K, its simplicial chain complex $C_\bullet(K)$ is defined as follows:

- $C_p(K)$ is the free abelian group generated by all p-simplices in K.
- The boundary homomorphism $\partial_p : C_p(K) \to C_{p-1}(K)$ is defined on the generator $[v_0, \ldots, v_p]$ by:

$$\partial_p([v_0, \ldots, v_p]) = \sum_{i=0}^{p} (-1)^i [v_0, \ldots, \widehat{v_i}, \cdots, v_p].$$

Proposition 2.9. *The simplicial chain complex* $(C_\bullet(K), \partial_\bullet)$ *satisfies* $\partial_p \circ \partial_{p+1} = 0$.

Definition 2.10 (Filtered Complex). Let K be a simplicial complex. A filtration of K is a family of subcomplexes $\{K_t\}_{t\in\mathbb{R}}$ such that:

1. If $s \leqslant t$, then $K_s \subseteq K_t$.
2. There exist $t_{\min}, t_{\max} \in \mathbb{R}$ such that $K_t = \varnothing$ for $t < t_{\min}$ and $K_t = K$ for $t \geqslant t_{\max}$.

The pair $(K, \{K_t\}_{t\in\mathbb{R}})$ is called a filtered complex.

Definition 2.11 (Persistent Homology). Let $\{K_t\}_{t\in\mathbb{R}}$ be a filtered complex. For each $t \in \mathbb{R}$, consider the simplicial chain complex $C_\bullet(K_t)$ and its homology groups $H_p(K_t)$.

For $s \leqslant t$, the inclusion map $i_{s,t} : K_s \hookrightarrow K_t$ induces a chain map $C_\bullet(K_s) \to C_\bullet(K_t)$, which in turn induces a homomorphism of homology groups:

$$\varphi_s^{t,(p)} \colon H_p(K_s) \to H_p(K_t).$$

The p-th persistent homology refers to the family of homology groups together with the transition homomorphisms between them:

$$\mathrm{PH}_p(\{K_t\}_{t\in\mathbb{R}}) = \left(H_p(K_t), \varphi_s^{t,(p)}\right)_{s\leqslant t\in\mathbb{R}}.$$

Definition 2.12 (Persistence Module). Regard $(\mathbb{R}, \leqslant)$ as a poset category. A persistence module M is a functor:

$$M \colon (\mathbb{R}, \leqslant) \to \mathsf{Vect}_K$$

where Vect_K is the category of vector spaces over a field K. Concretely:

1. For each $t \in \mathbb{R}$, a vector space M_t is assigned.
2. For each pair $s \leqslant t$, a linear map $\varphi_s^t \colon M_s \to M_t$ is assigned.
3. It satisfies $\varphi_t^t = \mathrm{id}_{M_t}$, and for $r \leqslant s \leqslant t$ we have $\varphi_r^t = \varphi_s^t \circ \varphi_r^s$.

Proposition 2.13. *Persistent homology* $\mathrm{PH}_p(\{K_t\})$ *naturally forms a persistence module.*

Proof. Define a functor $M : (\mathbb{R}, \leqslant) \to \mathsf{Vect}_K$ as follows:

1. For any $t \in \mathbb{R}$, set $M(t) = H_p(K_t)$.
2. For any $s \leqslant t$, define the transition map $M(s \leqslant t) = \varphi_s^{t,(p)} : H_p(K_s) \to H_p(K_t)$.

For any $t \in \mathbb{R}$, consider the inclusion map $i_{t,t} : K_t \hookrightarrow K_t$, which is the identity map. By basic properties of algebraic topology, the identity map induces the identity homomorphism on homology: $\varphi_t^{t,(p)} = \mathrm{id}_{H_p(K_t)}$, so $M(t \leqslant t) = \mathrm{id}_{M(t)}$.

For any $r \leqslant s \leqslant t$, consider the composition of inclusions:

$$K_r \xhookrightarrow{i_{r,s}} K_s \xhookrightarrow{i_{s,t}} K_t$$

By the definition of a filtration of simplicial complexes, this composition equals the inclusion $i_{r,t} : K_r \hookrightarrow K_t$.

At the chain complex level, these inclusions induce chain maps:

$$C_\bullet(K_r) \xrightarrow{(i_{r,s})_\sharp} C_\bullet(K_s) \xrightarrow{(i_{s,t})_\sharp} C_\bullet(K_t)$$

and $(i_{s,t})_\sharp \circ (i_{r,s})_\sharp = (i_{r,t})_\sharp$.

At the homology level, chain maps induce homomorphisms:

$$H_p(K_r) \xrightarrow{\varphi_r^{s,(p)}} H_p(K_s) \xrightarrow{\varphi_s^{t,(p)}} H_p(K_t)$$

By the functoriality of homology, we have $\varphi_s^{t,(p)} \circ \varphi_r^{s,(p)} = \varphi_r^{t,(p)}$, i.e., $M(s \leqslant t) \circ M(r \leqslant s) = M(r \leqslant t)$. □

Definition 2.14 (Interval Module). Let $I \subseteq \mathbb{R}$ be an interval and K a field. The interval module $\mathbb{I}_I$ is defined by:

$$\mathbb{I}_I(t) := \begin{cases} K, & t \in I, \\ 0, & t \notin I. \end{cases}$$

The transition map $\varphi_s^t : \mathbb{I}_I(s) \to \mathbb{I}_I(t)$ is the identity map if $s, t \in I$, and the zero map otherwise.

Theorem 2.15 (Structure Theorem for Persistence Modules)**.** *Let K be a field and M a persistence module over $\mathbb{R}$. Then there exists a unique multiset* $\mathrm{Barc}(M)$ *of intervals (called barcodes) such that:*

$$M \cong \bigoplus_{I \in \mathrm{Barc}(M)} \mathbb{I}_I.$$

Definition 2.16 (Persistence Diagram). The persistence diagram $\mathrm{Dgm}(M)$ of a persistence module M is the graphical representation of its barcode $\mathrm{Barc}(M)$, where each interval $I = [b, d)$ is represented as a point $(b, d) \in \mathbb{R}^2$.

Definition 2.17 (Bottleneck Distance). Let $\mathrm{Dgm}_1, \mathrm{Dgm}_2$ be two persistence diagrams. Their bottleneck distance is defined as:

$$d_B(\mathrm{Dgm}_1, \mathrm{Dgm}_2) := \inf_\gamma \sup_{x \in \mathrm{Dgm}_1} \|x - \gamma(x)\|_\infty.$$

where γ ranges over all partial bijections between Dgm_1 and Dgm_2, and points may be matched to the diagonal $\Delta = \{(x, x) \in \mathbb{R}^2\}$.

Theorem 2.18 (Stability Theorem)**.** *Let X, Y be finite point sets in a compact metric space, and let $\{K_t(X)\}, \{K_t(Y)\}$ be the corresponding filtered complexes (such as Čech or Vietoris–Rips filtrations). Then*

$$d_B(\mathrm{PH}_p(X), \mathrm{PH}_p(Y)) \leqslant d_H(X, Y).$$

where d_H denotes the Hausdorff distance.

Definition 2.19 (Cohomology Group). Let K be a simplicial complex and G an abelian group. The cochain complex $C^\bullet(K;G)$ of K is defined by:

1. $C^p(K;G) := \mathrm{Hom}(C_p(K), G)$, i.e., the set of group homomorphisms from the p-chain group to G.

2. For any $(p+1)$-simplex, the coboundary operator $\delta^p : C^p(K;G) \to C^{p+1}(K;G)$ is defined by $(\delta^p f)(\sigma) = f(\partial_{p+1}\sigma)$.

The p-th cohomology group of K is defined as:

$$H^p(K;G) := {\ker \delta^p}\big/{\delta^{p-1}C^{p-1}(K;G)}$$

where elements of $\ker \delta^p$ are called p-cocycles, and elements of $\delta^{p-1}C^{p-1}(K;G)$ are called p-coboundaries.

Definition 2.20 (Persistent Cohomology). Let $\{K_t\}_{t\in\mathbb{R}}$ be a filtered complex. For each $t \in \mathbb{R}$, consider the cochain complex $C^\bullet(K_t;G)$ and its cohomology groups $H^p(K_t;G)$.

For $s \leqslant t$, the inclusion map $i_{s,t} : K_s \hookrightarrow K_t$ induces a chain map $C_\bullet(K_s) \to C_\bullet(K_t)$, which by duality induces a cochain map $C^\bullet(K_t;G) \to C^\bullet(K_s;G)$, and finally a homomorphism of cohomology groups:

$$\psi_s^{t,(p)} \colon H^p(K_t;G) \to H^p(K_s;G).$$

The p-th persistent cohomology refers to the family of cohomology groups together with the transition homomorphisms between them:

$$\mathrm{PH}^p(\{K_t\}_{t\in\mathbb{R}};G) := \left(H^p(K_t;G), \psi_s^{t,(p)}\right)_{s\leqslant t\in\mathbb{R}}$$

where $\psi_s^{t,(p)} : H^p(K_t) \to H^p(K_s)$ is the pullback homomorphism on cohomology induced by the inclusion $K_s \hookrightarrow K_t$.

Definition 2.21 (Nerve of an Open Cover). Let X be a topological space and $\mathscr{U} = \{U_i\}_{i\in I}$ an open cover of X. The nerve of the cover $\mathscr{U}$ is the abstract simplicial complex $\mathcal{N}(\mathscr{U})$ defined by:

1. Vertex set: $V = \{v_i : i \in I\}$, where each vertex v_i corresponds to the open set U_i.

2. Simplices: A finite subset $\{v_{i_0}, v_{i_1}, \ldots, v_{i_k}\} \subseteq V$ forms a k-simplex if and only if the intersection of the corresponding open sets is nonempty:

$$U_{i_0} \cap U_{i_1} \cap \cdots \cap U_{i_k} \neq \varnothing.$$

Lemma 2.22 (Nerve Lemma). *Let $\mathscr{U} = \{U_i\}_{i\in I}$ be an open cover of a topological space X. If every nonempty finite intersection $U_{i_0} \cap U_{i_1} \cap \cdots \cap U_{i_k}$ is contractible (i.e., homotopy equivalent to a point), then the geometric realization of $\mathcal{N}(\mathscr{U})$ is homotopy equivalent to X:*

$$|\mathcal{N}(\mathscr{U})| \simeq X.$$

Lemma 2.23 (Čech Cover Approximation Lemma). *Let K be a compact subset of $\mathbb{R}^D$, let $\mathscr{U} = \{B(z,\varepsilon) \mid z \in K\}$ be the ε-ball cover of K, and let $\mathscr{V} = \{B(z_n,\varepsilon) \mid z_n \in Z\}$ be a "sample subcover" of $\mathscr{U}$ with $Z \subset K$. If Z is ρ-dense in K, i.e., $\forall z \in K$, $\exists z_n \in Z$ such that $d(z,z_n) < \rho$, and $\rho < \varepsilon/2$, then:*

1. *$\mathscr{V}$ is an open cover of K;*

2. *There exist natural simplicial maps $f : \mathcal{N}(\mathscr{U}) \to \mathcal{N}(\mathscr{V})$ and $g : \mathcal{N}(\mathscr{V}) \to \mathcal{N}(\mathscr{U})$ such that $g \circ f \simeq id_{\mathcal{N}(\mathscr{U})}$ and $f \circ g \simeq id_{\mathcal{N}(\mathscr{V})}$, i.e., $|\mathcal{N}(\mathscr{U})| \simeq |\mathcal{N}(\mathscr{V})|$.*

2.2 Čech Cohomology and Sheaves

Definition 2.24 (Presheaf). A presheaf $\mathscr{F}$ on a topological space X assigns to each nonempty open set U of X a set[1] $\mathscr{F}(U)$, and to any inclusion of open sets $V \subseteq U$ a homomorphism, called the restriction map $r_V^U \colon \mathscr{F}(U) \to \mathscr{F}(V)$, such that

1. $r_U^U = \mathrm{id}$.
2. For any $W \subseteq V \subseteq U$, we have $r_W^U = r_W^V \circ r_V^U$.

Elements of $\mathscr{F}(U)$ are called sections of $\mathscr{F}$ over U.

Let $\mathscr{F}, \mathscr{G}$ be presheaves on X. A morphism h from $\mathscr{F}$ to $\mathscr{G}$ assigns to each nonempty open set U a homomorphism $h_U \colon \mathscr{F}(U) \to \mathscr{G}(U)$ such that for any $V \subseteq U$, the following diagram commutes:

$$\begin{tikzcd} \mathscr{F}(U) \arrow[r, "h_U"] \arrow[d, "r_V^U"] & \mathscr{G}(U) \arrow[d, "\rho_V^U"] \\ \mathscr{F}(V) \arrow[r, "h_V"] & \mathscr{G}(V) \end{tikzcd}$$

Definition 2.25 (Sheaf). A presheaf $\mathscr{F}$ is called a sheaf if for any open set U and any open cover $\mathscr{U} = \bigcup_{i \in I} U_i$ of U, the following hold:

1. If $s, t \in \mathscr{F}(U)$ satisfy $r_{U_i}^U(s) = r_{U_i}^U(t)$ for all $i \in I$, then $s = t$.
2. If $s_i \in \mathscr{F}(U_i)$, $s_j \in \mathscr{F}(U_j)$ satisfy $r_{U_i \cap U_j}^{U_i}(s_i) = r_{U_i \cap U_j}^{U_j}(s_j)$ whenever $U_i \cap U_j \neq \varnothing$, then there exists $s \in \mathscr{F}(U)$ such that $r_{U_i}^U(s) = s_i$ for all $i \in I$.

Definition 2.26 (Sheafification). For any $p \in X$, the stalk $\mathscr{F}_p$ of a presheaf $\mathscr{F}$ on X at p is defined by

$$\mathscr{F}_p := \varinjlim_{U \ni p} \mathscr{F}(U) := \coprod_{U \ni p} \mathscr{F}(U) \Big/ \sim$$

where $s \in \mathscr{F}(U) \sim t \in \mathscr{F}(V)$ if and only if there exists an open set $W \subseteq U \cap V$ such that $r_W^U(s) = r_W^V(t)$.

Elements of $\mathscr{F}_p$ are called germs.

For any $U \ni p$, define the map $\mathscr{F}(U) \to \mathscr{F}_p, \quad f \mapsto f_p := [f]$.

The sheafification of a presheaf $\mathscr{F}$ is defined by

$$\mathscr{F}^{\#}(U) := \left\{ s \colon U \to \coprod_{p \in U} \mathscr{F}_p \;\middle|\; \begin{array}{l} \forall q \in U, \exists \text{ an open neighborhood } V \subseteq U \text{ and a section} \\ f \in \mathscr{F}(V) \text{ such that } s(\xi) = f_\xi \text{ for all } \xi \in V \end{array} \right\}$$

with restriction homomorphisms $r_V^U := s|_V$. Then $\mathscr{F}^{\#}$ is a sheaf.

Naturally, one defines a morphism $\theta \colon \mathscr{F} \to \mathscr{F}^{\#}$ from the presheaf to its sheafification, such that for any open set U,

$$\theta_U \colon \mathscr{F}(U) \to \mathscr{F}^{\#}(U), \quad s \mapsto \left(\begin{array}{ccc} \widetilde{s} \colon U & \longrightarrow & \coprod_{p \in U} \mathscr{F}_p \\ p & \longmapsto & s_p \end{array} \right)$$

[1] More generally, an abelian group, a ring, etc.

Remark 2.27. 1. If $\mathscr{F}$ is a sheaf, then θ is an isomorphism.

2. Let $\mathscr{G}$ be a sheaf. For any morphism $h\colon \mathscr{F} \to \mathscr{G}$, there exists a unique $\widetilde{h}\colon \mathscr{F}^{\#} \to \mathscr{G}$ such that the following diagram commutes.

$$\begin{tikzcd} \mathscr{F} \arrow[rr, "h"] \arrow[dr, "\theta"'] & & \mathscr{G} \\ & \mathscr{F}^{\#} \arrow[ur, dashed, "\widetilde{h}"'] & \end{tikzcd}$$

Definition 2.28 (Čech Cohomology of a Cover). Let $\mathscr{F}$ be a sheaf on X, and let $\mathscr{U} = \{U_i\}_{i\in I}$ be an open cover of X. For a multi-index $(i_0, \dots, i_p) \in I^{p+1}$, write $U_{i_0,\dots,i_p} := U_{i_0} \cap \cdots \cap U_{i_p}$. Define

$$C^p(\mathscr{U}, \mathscr{F}) := \prod_{(i_0,\cdots,i_p)\in I^{p+1}} \mathscr{F}(U_{i_0,\dots,i_p}),$$

concretely, for $\sigma \in C^p(\mathscr{U}, \mathscr{F})$, we have $\sigma = \{\sigma_{i_0,\dots,i_p} \colon \sigma_{i_0,\dots,i_p} \in \mathscr{F}(U_{i_0,\dots,i_p})\}$.

Define the coboundary operator $\delta\colon C^p(\mathscr{U}, \mathscr{F}) \to C^{p+1}(\mathscr{U}, \mathscr{F}), \quad \sigma \mapsto \delta\sigma$, where

$$(\delta\sigma)_{i_0,\dots,i_{p+1}} := \sum_{j=0}^{p+1} (-1)^j r^{U_{i_0,\dots,\widehat{i_j},\dots,i_{p+1}}}_{U_{i_0,\dots,i_{p+1}}} (\sigma_{i_0,\dots,\widehat{i_j},\dots,i_{p+1}}) =: \sum_{j=0}^{p+1} (-1)^j \sigma_{i_0,\dots,\widehat{i_j},\dots,i_{p+1}}|_{U_{i_0,\dots,i_{p+1}}}.$$

Set $Z^p(\mathscr{U}, \mathscr{F}) := \ker\delta \subseteq C^p(\mathscr{U}, \mathscr{F})$, $B^p(\mathscr{U}, \mathscr{F}) := \delta C^{p-1}(\mathscr{U}, \mathscr{F})$. Then the Čech cohomology groups of the cover $\mathscr{U}$ are defined by

$$\check{H}^p(\mathscr{U}; \mathscr{F}) := {Z^p(\mathscr{U}, \mathscr{F})}/{B^p(\mathscr{U}, \mathscr{F})}.$$

If $\mathscr{V} = \{V_\alpha\}_{\alpha\in\Gamma}$ is a refinement of $\mathscr{U} = \{U_i\}_{i\in I}$, choose a map $\tau\colon \Gamma \to I$ such that $V_\alpha \subseteq U_{\tau(\alpha)}$. Then one defines a homomorphism to the refinement

$$\Phi^{\mathscr{U}}_{\mathscr{V}} : \check{H}^p(\mathscr{U}; \mathscr{F}) \to \check{H}^p(\mathscr{V}; \mathscr{F}), \quad [\sigma] = [\{\sigma_{i_0,\dots,i_p}\}] \mapsto [\eta] = [\{\eta_{\alpha_0,\dots,\alpha_p}\}]$$

where $\eta_{\alpha_0,\dots,\alpha_p} \in \mathscr{F}(V_{\alpha_0,\dots,\alpha_p}) \subseteq \mathscr{F}(U_{\tau(\alpha_0),\dots,\tau(\alpha_p)})$.

Definition 2.29 (Čech Cohomology of a Space). Furthermore, the Čech cohomology groups of a space X are defined by

$$\check{H}^p(X; \mathscr{F}) := \varinjlim_{\mathscr{U}} \check{H}^p(\mathscr{U}; \mathscr{F}) := \coprod_{\mathscr{U}} \check{H}^p(\mathscr{U}; \mathscr{F}) \Big/ \sim$$

where the equivalence relation is given by:

$$[\sigma] \in \check{H}^p(\mathscr{U}; \mathscr{F}) \sim [\eta] \in \check{H}^p(\mathscr{V}; \mathscr{F}) \iff \exists \text{ a common refinement } \mathscr{W} \text{ such that } \Phi^{\mathscr{U}}_{\mathscr{W}}([\sigma]) = \Phi^{\mathscr{V}}_{\mathscr{W}}([\eta]).$$

A sequence of sheaves is called exact if the induced sequence on stalks is exact at every point.

Definition 2.30 (Resolution). Let $\mathscr{F}$ be a sheaf on a topological space X. A resolution of $\mathscr{F}$ is an exact sequence

$$0 \to \mathscr{F} \to \mathscr{S}^0 \to \mathscr{S}^1 \to \mathscr{S}^2 \to \cdots$$

where each $\mathscr{S}^p$ is a sheaf on X.

Definition 2.31 (Acyclic Resolution). Let $\mathscr{F}$ be a sheaf on a topological space X. A resolution $0 \to \mathscr{F} \to \mathscr{S}^\bullet$ is called acyclic if each $\mathscr{S}^p$ is acyclic, i.e., $\check{H}^i(X; \mathscr{S}^p) = 0$ for all $i > 0$.

Definition 2.32 (Fine Sheaf). Let $\mathscr{F}$ be a sheaf on a topological space X. $\mathscr{F}$ is called fine if for any locally finite open cover $\{U_i\}$ of X, there exists a partition of unity $\{e_i\}$ subordinate to the cover, i.e.,

1. $\operatorname{supp}(e_i) \subseteq U_i$.
2. $\sum_i e_i = 1$.

Theorem 2.33 (Abstract de Rham Theorem). *Let $\mathscr{F}$ be a sheaf on a topological space X, and let $0 \to \mathscr{F} \to \mathscr{S}^\bullet$ be an acyclic resolution of $\mathscr{F}$ such that each $\mathscr{S}^p$ is a fine sheaf. Then the Čech cohomology $\check{H}^p(X, \mathscr{F})$ of $\mathscr{F}$ is isomorphic to the p-th cohomology of the complex $\Gamma(X, \mathscr{S}^\bullet)$:*

$$\check{H}^p(X, \mathscr{F}) \cong \ker\left(\Gamma(X, \mathscr{S}^p) \to \Gamma(X, \mathscr{S}^{p+1})\right) \Big/ \operatorname{im}\left(\Gamma(X, \mathscr{S}^{p-1}) \to \Gamma(X, \mathscr{S}^p)\right).$$

In particular, if $\mathscr{U}$ is an open cover of X such that for every finite nonempty intersection $U_{i_0,\ldots,i_k}$ of sets in $\mathscr{U}$, we have $\check{H}^j(U_{i_0,\ldots,i_k}; \mathscr{S}^p) = 0$ for all $j > 0$, then there is a natural isomorphism

$$\check{H}^p(\mathscr{U}; \mathscr{F}) \cong \check{H}^p(X, \mathscr{F}).$$

3 Dynamic Čech Complexes and a Sheaf-Theoretic Unified Framework

3.1 Theory of Dynamic Čech Complexes

When dealing with dynamical systems, traditional TDA typically uses a discrete approach: select a discrete sequence $\{t_1, \ldots, t_m\}$ on the time axis, construct a static Čech complex at each time t_i, then compare topological features among these discrete complexes. However, this method is essentially a discrete approximation and cannot describe the continuous evolution of topological structures over the parameter space. In this subsection, we propose a fundamental viewpoint: model the dynamical process as a continuous map from the parameter space T to the space of complexes $\mathcal{S}(M)$, thereby establishing a truly continuous dynamical theory.

Definition 3.1 (Continuous Complex Flow). Let the parameter space T be a connected topological space, and let (M, g) be a compact d-dimensional Riemannian manifold. A *continuous complex flow* is a continuous map:

$$\mathscr{C} \colon T \to \mathcal{S}(M)$$

where $\mathcal{S}(M)$ denotes the topological space of all abstract simplicial complexes on M, endowed with the Whitney topology. That is, for any open set $U \subseteq M$ and integer $k \geqslant 0$, a subbase of open sets is given by

$$\mathcal{U}(U, k) := \{K \in \mathcal{S}(M) \mid K \cap U \text{ contains at least one } k\text{-simplex}\}.$$

The core idea is to model the dynamical process as a "path" in the parameter space, whose continuous movement in $\mathcal{S}(M)$ describes the continuous evolution of topological structure.

Definition 3.2 (Parametrized Topological Space). Let the parameter space T be a connected topological space, and let M be a compact d-dimensional manifold. A *parametrized topological space* is a trivial bundle $\pi \colon E = M \times T \to T$ satisfying fiber isomorphism $\pi^{-1}(t) \cong M$.

Definition 3.3 (Dynamic Family of Embeddings). A *dynamic family of embeddings* is a smooth map $\Phi\colon E = M \times T \to \mathbb{R}^D$ such that

1. $\Phi_t := \Phi(\cdot, t)\colon M \to \mathbb{R}^D$ is an embedding;
2. There exists $c > 0$ such that, for every $t \in T$, the norm of the Jacobian matrix of Φ_t satisfies $\|\mathbf{J}\Phi_t\| \leqslant c$ for all $x \in M$;
3. The map $t \mapsto \Phi_t$ is C^∞.

Within the abstract framework of continuous complex flows, we require concrete geometric realizations. The dynamic Čech continuous flow serves as precisely such a bridge.

Definition 3.4 (Dynamic Čech Continuous Flow). Given a dynamic family of embeddings $\Phi\colon M \times T \to \mathbb{R}^D$ and a radius parameter $\varepsilon > 0$, the *dynamic Čech continuous flow* is defined by

$$\begin{aligned} \mathscr{C}_{\Phi,\varepsilon}\colon T &\longrightarrow \mathcal{S}(M) \\ t &\longmapsto \Big\{\sigma \subseteq M \mid \bigcap_{x\in\sigma} B(\Phi_t(x), \varepsilon) \neq \varnothing\Big\} \end{aligned}$$

Theorem 3.5 (Continuity Guarantee)**.** *If the dynamic family of embeddings Φ is continuous, then the dynamic Čech continuous flow $\mathscr{C}_{\Phi,\varepsilon}$ is continuous with respect to the Whitney topology.*

Proof. Note that the Whitney topology on $\mathcal{S}(M)$ is generated by the sets $\mathcal{U}(U, k)$. Thus to prove continuity of $\mathscr{C}_{\Phi,\varepsilon}$, it suffices to show that for every $\mathcal{U}(U, k)$, its preimage

$$\mathscr{C}_{\Phi,\varepsilon}^{-1}\big(\mathcal{U}(U, k)\big) := \big\{t \in T \mid \mathscr{C}_{\Phi,\varepsilon}(t) \in \mathcal{U}(U, k)\big\}$$

is open in T.

Let $t_0 \in \mathscr{C}_{\Phi,\varepsilon}^{-1}(\mathcal{U}(U, k))$. Then there exists a k-simplex $\sigma = \{x_1, \ldots, x_{k+1}\} \subseteq U$ in $\mathscr{C}_{\Phi,\varepsilon}(t_0)$, so there exists $y \in \mathbb{R}^D$ such that $y \in \bigcap_{i=1}^{k+1} B(\Phi_{t_0}(x_i), \varepsilon)$, i.e., $d(\Phi_{t_0}(x_i), y) < \varepsilon$ for all i.

By continuity of Φ, for each x_i, the map $t \mapsto \Phi_t(x_i)$ is continuous at t_0. Hence there exists $\delta_i > 0$ such that whenever $d_T(t, t_0) < \delta_i$,

$$d_{\mathbb{R}^D}(\Phi_t(x_i), \Phi_{t_0}(x_i)) < \eta, \quad \eta = \frac{1}{2}\big(\varepsilon - \max_i d(\Phi_{t_0}(x_i), y)\big) > 0.$$

Set $\delta = \min\{\delta_i \mid 1 \leqslant i \leqslant k+1\}$. Then for $d_T(t, t_0) < \delta$, for every i,

$$d(\Phi_t(x_i), y) \leqslant d(\Phi_t(x_i), \Phi_{t_0}(x_i)) + d(\Phi_{t_0}(x_i), y) < \eta + d(\Phi_{t_0}(x_i), y) < \varepsilon.$$

Thus $y \in \bigcap_{i=1}^{k+1} B(\Phi_t(x_i), \varepsilon)$, so $\sigma \in \mathscr{C}_{\Phi,\varepsilon}(t)$. Since $\sigma \subseteq U$, we have $\mathscr{C}_{\Phi,\varepsilon}(t) \in \mathcal{U}(U, k)$, so $B_{d_T}(t_0, \delta) \subseteq \mathscr{C}_{\Phi,\varepsilon}^{-1}(\mathcal{U}(U, k))$. This shows that the preimage is open. □

This result shows that continuity of the parameter space naturally descends to continuity of the complex structure. In what follows, we will gradually see that this preservation of continuity has profound implications.

To understand dynamical topological processes at a more abstract level, we lift the concept of continuous complex flow to a categorical setting. This lifting is not merely a formal nicety, but reveals the inherent algebraic structure of dynamical topological processes, providing a natural framework for subsequent sheaf-theoretic constructions.

In traditional static TDA, a Čech complex is viewed as an object in the category Simp. While concise, this viewpoint cannot capture the dynamical behavior of complexes varying with

parameters. To describe such dynamics, we elevate our perspective from "objects" to "functors". Thus we uniformly define a dynamic Čech complex as a functor from the parameter space to the category of simplicial complexes, a construction that simultaneously encodes the continuous evolution of topology and its algebraic structure.

Definition 3.6 (Dynamic Čech Complex Functor). Let (M, g) be a compact d-dimensional Riemannian manifold, and let the parameter space T be a connected topological space. Given a dynamic family of embeddings $\Phi\colon M \times T \to \mathbb{R}^D$ and a radius parameter $\varepsilon > 0$, the *dynamic Čech complex functor* is defined by:

$$\mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon} : T \to \mathsf{Simp}$$

where:

1. For any $t \in T$, $\mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(t) := \{\sigma \subseteq M \mid \bigcap_{x\in\sigma} B(\Phi_t(x), \varepsilon) \neq \varnothing\}$.
2. For any $s, t \in T$ and path $\gamma : s \to t$, i.e., $\gamma\colon [0,1] \to T$ with $\gamma(0) = s$, $\gamma(1) = t$, $\mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(\gamma) : \mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(s) \to \mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(t)$ is the induced simplicial map.

This unified construction has a dual significance:

- From the continuous flow viewpoint, regarding $\mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}$ as a map from T to the complex space $\mathcal{S}(M)$, it describes the continuous evolution of topological structure, capturing the infinitesimal changes of topological features in the dynamical process (see later sections).
- From the functorial viewpoint, regarding $\mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}$ as a functor from the parameter category T to the simplicial complex category Simp, it reveals the algebraic structure of the dynamical process, providing a natural framework for subsequent sheaf and cohomology theories.

In practical applications, we typically only have access to a finite sample set of the manifold M. This motivates the sampled version of the definition:

Definition 3.7 (Sampled Dynamic Čech Complex Functor). Let $X_n \subset M$ be a finite sample set. The *sampled dynamic Čech complex functor* is defined by:

$$\mathscr{C}^{\mathrm{smp}}_{\Phi,\varepsilon} : T \to \mathsf{Simp}$$

where for any $t \in T$,

$$\mathscr{C}^{\mathrm{smp}}_{\Phi,\varepsilon}(t) := \left\{ \sigma \subseteq X_n \,\middle|\, \bigcap_{x\in\sigma} B(\Phi_t(x), \varepsilon) \neq \varnothing \right\},$$

and for any path $\gamma : s \to t$, $\mathscr{C}^{\mathrm{smp}}_{\Phi,\varepsilon}(\gamma)$ is the corresponding simplicial map.

Proposition 3.8 (Restriction Relation)**.** *The sampled dynamic Čech complex functor is the restriction of the full dynamic Čech complex functor to the sample set X_n:*

$$\mathscr{C}^{\mathrm{smp}}_{\Phi,\varepsilon} = \mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}\Big|_{X_n}.$$

That is, for any $t \in T$, $\mathscr{C}^{\mathrm{smp}}_{\Phi,\varepsilon}(t)$ is the subcomplex of $\mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(t)$ generated by the sample set X_n.

Proof. This follows directly from the definition: the simplices of $\mathscr{C}^{\mathrm{smp}}_{\Phi,\varepsilon}(t)$ are exactly those subsets of simplices in $\mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(t)$ whose vertex set is contained in X_n. □

Theorem 3.9 (Approximation Property)**.** *If the sample set X_n is δ-dense in M, i.e., for every $x \in M$, there exists $x_i \in X_n$ such that $d_M(x, x_i) < \delta$, then as $\delta \to 0$, the sampled dynamic Čech complex functor converges to the full dynamic Čech complex functor in an appropriate topological sense:*

$$\mathscr{C}^{\mathrm{smp}}_{\Phi,\varepsilon} \to \mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon} \quad (\delta \to 0).$$

In particular, for any $t \in T$, there is a homotopy equivalence

$$\left|\mathscr{C}^{\mathrm{smp}}_{\Phi,\varepsilon}(t)\right| \simeq \left|\mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(t)\right|, \quad \textit{for sufficiently small } \delta.$$

Proof. Fix $t \in T$ and consider $\Phi_t : M \to \mathbb{R}^D$. For any $x, y \in M$, since M is a compact Riemannian manifold, there exists a smooth curve $\gamma : [0,1] \to M$ connecting x and y with $\gamma(0) = x$, $\gamma(1) = y$. Then

$$\Phi_t(y) - \Phi_t(x) = \int_0^1 \mathbf{J}\Phi_t(\gamma(s)) \cdot \gamma'(s)\,\mathrm{d}s.$$

Taking the Euclidean norm,

$$d_{\mathbb{R}^D}(\Phi_t(x), \Phi_t(y)) \leqslant \int_0^1 \|\mathbf{J}\Phi_t(\gamma(s))\| \cdot \|\gamma'(s)\|\,\mathrm{d}s \leqslant c \int_0^1 \|\gamma'(s)\|\,\mathrm{d}s = c \cdot d_M(x, y).$$

Hence Φ_t is Lipschitz continuous with Lipschitz constant c.

If $X_n \subset M$ is δ-dense, then for any $x \in M$ and $t \in T$,

$$d_{\mathbb{R}^D}(\Phi_t(x), \Phi_t(x_n)) \leqslant c \cdot d_M(x, x_n) < c\delta.$$

That is, $\Phi_t(X_n)$ is $c\delta$-dense in $\Phi_t(M)$.

Fix $t \in T$ and consider the following two open covers:

1. The full cover $\mathscr{U}_t := \{B(\Phi_t(x), \varepsilon) \mid x \in M\}$, whose nerve is $\mathcal{N}(\mathscr{U}_t) = \mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(t)$.
2. The sampled cover $\mathscr{U}_t^{\mathrm{smp}} := \{B(\Phi_t(x), \varepsilon) \mid x \in X_n\}$, whose nerve is $\mathcal{N}(\mathscr{U}_t^{\mathrm{smp}}) = \mathscr{C}^{\mathrm{smp}}_{\Phi,\varepsilon}(t)$.

We show that if $\delta < \varepsilon/(2c)$, then $|\mathcal{N}(\mathscr{U}_t)| \simeq |\mathcal{N}(\mathscr{U}_t^{\mathrm{smp}})|$.

Define a map $f : |\mathcal{N}(\mathscr{U}_t)| \to |\mathcal{N}(\mathscr{U}_t^{\mathrm{smp}})|$ as follows. For any point $y \in |\mathcal{N}(\mathscr{U}_t)|$, write its barycentric coordinate representation as $y = \sum_{x \in \sigma} \lambda_x \cdot v_x$, where $\sigma = \{x_0, \ldots, x_k\} \in \mathcal{N}(\mathscr{U}_t)$, so $\bigcap_{x \in \sigma} B(\Phi_t(x), \varepsilon) \neq \varnothing$, $\lambda_x \geqslant 0$, $\sum \lambda_x = 1$, and v_x is the vertex corresponding to x.

By δ-density of X_n, for each $x \in \sigma$, choose $\pi(x) \in X_n$ such that $d_M(x, \pi(x)) < \delta$. Define $f(y) = \sum_{x \in \sigma} \lambda_x \cdot v_{\pi(x)}$, where $v_{\pi(x)}$ is the vertex in $\mathcal{N}(\mathscr{U}_t^{\mathrm{smp}})$ corresponding to $\pi(x)$.

We verify that f is well-defined, i.e., that $\{\pi(x) \mid x \in \sigma\}$ forms a simplex in $\mathcal{N}(\mathscr{U}_t^{\mathrm{smp}})$. Since $\sigma \in \mathcal{N}(\mathscr{U}_t)$, there exists $z \in \bigcap_{x \in \sigma} B(\Phi_t(x), \varepsilon)$, so $d_{\mathbb{R}^D}(\Phi_t(x), z) < \varepsilon$. For any $x \in \sigma$,

$$d_{\mathbb{R}^D}(\Phi_t(\pi(x)), z) \leqslant d_{\mathbb{R}^D}(\Phi_t(\pi(x)), \Phi_t(x)) + d_{\mathbb{R}^D}(\Phi_t(x), z) < c\delta + \varepsilon.$$

Since $\delta < \varepsilon/(2c)$, we have $d_{\mathbb{R}^D}(\Phi_t(\pi(x)), z) < 3\varepsilon/2$. Thus $\bigcap_{\pi(x) \in \pi(\sigma)} B(\Phi_t(\pi(x)), 3\varepsilon/2) \neq \varnothing$. Combined with classical Čech approximation theory, for sufficiently small δ, $\pi(\sigma)$ is a simplex in $\mathcal{N}(\mathscr{U}_t^{\mathrm{smp}})$.

Define a map $g : |\mathcal{N}(\mathscr{U}_t^{\mathrm{smp}})| \to |\mathcal{N}(\mathscr{U}_t)|$ as follows. For each vertex v_{x_n} in the sampled nerve ($x_n \in X_n$), set $g(v_{x_n}) = v_{x_n}$ and extend linearly over simplices.

We obtain a homotopy equivalence:

1. $g \circ f \sim \mathrm{id}_{|\mathcal{N}(\mathscr{U}_t)|}$: construct a homotopy $H(s, y) = \sum \lambda_x \cdot (sv_x + (1-s)v_{\pi(x)})$. Since x and $\pi(x)$ are sufficiently close, the simplex persists throughout the deformation.

2. $f \circ g \sim \mathrm{id}_{|\mathcal{N}(\mathscr{U}_t^{\mathrm{smp}})|}$: similar homotopy, using density of the sample to ensure continuity.

Hence $|\mathcal{N}(\mathscr{U}_t)| \simeq |\mathcal{N}(\mathscr{U}_t^{\mathrm{smp}})|$.

Therefore, as $\delta \to 0$, $\mathscr{C}_{\Phi,\varepsilon}^{\mathrm{smp}} \to \mathscr{C}_{\Phi,\varepsilon}^{\mathrm{dyn}}$, and for any $t \in T$, $|\mathscr{C}_{\Phi,\varepsilon}^{\mathrm{smp}}(t)| \simeq |\mathscr{C}_{\Phi,\varepsilon}^{\mathrm{dyn}}(t)|$. □

Theorem 3.10 (Continuity Guarantee)**.** *If the dynamic family of embeddings Φ is continuous, then the dynamic Čech complex functor $\mathscr{C}_{\Phi,\varepsilon}^{\mathrm{dyn}}$ and the sampled dynamic Čech complex functor $\mathscr{C}_{\Phi,\varepsilon}^{\mathrm{smp}}$ are continuous in the following sense:*

1. *As maps from T to $\mathcal{S}(M)$, they are continuous in the Whitney topology.*

2. *As functors from T to* Simp*, they preserve the path structure of the parameter space.*

Proof. The proof of topological continuity is analogous to earlier arguments. We prove functorial continuity; the sampled version is similar.

We verify that $\mathscr{C}_{\Phi,\varepsilon}^{\mathrm{dyn}}$ satisfies the functor axioms. Preservation of identities is clear: for any $t \in T$, the constant path $\mathrm{id}_t : t \to t$ induces the identity simplicial map:

$$\mathscr{C}_{\Phi,\varepsilon}^{\mathrm{dyn}}(\mathrm{id}_t) = \mathrm{id}_{\mathscr{C}_{\Phi,\varepsilon}^{\mathrm{dyn}}(t)}.$$

Let $\gamma : s \to t$ and $\delta : t \to u$ be paths in T (assumed at least C^1). That is, $\gamma : [0,1] \to T$ with $\gamma(0) = s$, $\gamma(1) = t$, and $\delta : [0,1] \to T$ with $\delta(0) = t$, $\delta(1) = u$. Define the composite path $\delta \circ \gamma : [0,1] \to T$ by

$$(\delta \circ \gamma)(\tau) = \begin{cases} \gamma(2\tau), & \tau \in [0, 1/2], \\ \delta(2\tau - 1), & \tau \in (1/2, 1]. \end{cases}$$

We show composition is preserved:

$$\mathscr{C}_{\Phi,\varepsilon}^{\mathrm{dyn}}(\delta \circ \gamma) = \mathscr{C}_{\Phi,\varepsilon}^{\mathrm{dyn}}(\delta) \circ \mathscr{C}_{\Phi,\varepsilon}^{\mathrm{dyn}}(\gamma).$$

For the path $\gamma : s \to t$, consider the simplicial map $\mathscr{C}_{\Phi,\varepsilon}^{\mathrm{dyn}}(\gamma) : \mathscr{C}_{\Phi,\varepsilon}^{\mathrm{dyn}}(s) \to \mathscr{C}_{\Phi,\varepsilon}^{\mathrm{dyn}}(t)$. For each vertex $x \in M$, since Φ is C^∞,

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\Phi_{\gamma(\tau)}(x) = \frac{\partial \Phi}{\partial t}(x, \gamma(\tau)) \cdot \gamma'(\tau).$$

With initial condition $\Phi_{\gamma(0)}(x) = \Phi_s(x)$, there is a unique solution $\Phi_{\gamma(\tau)}(x)$, depending continuously on τ.

Thus for each vertex $x \in \mathscr{C}_{\Phi,\varepsilon}^{\mathrm{dyn}}(s)$, we define its image to be x itself (the vertex set is unchanged), giving the identity map on vertices.

For a simplex $\sigma = \{x_1, \ldots, x_k\} \in \mathscr{C}_{\Phi,\varepsilon}^{\mathrm{dyn}}(s)$, we determine its image. Since the vertex map is the identity, the vertex set of σ maps to $\{x_1, \ldots, x_k\}$. However, $\{x_1, \ldots, x_k\}$ may no longer be a simplex at time t.

We define: the image of σ under the simplicial map is $\{x_1, \ldots, x_k\}$ if and only if σ remains a simplex along the entire path γ, i.e.,

$$\bigcap_{i=1}^{k} B(\Phi_{\gamma(\tau)}(x_i), \varepsilon) \neq \varnothing, \quad \forall \tau \in [0,1].$$

If this holds, set $\mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(\gamma)(\sigma) = \sigma$; otherwise, take the maximal subsimplex $\tau \subset \sigma$ that remains a simplex along γ, and set $\mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(\gamma)(\sigma) = \tau$.

We now prove composition is preserved. For any simplex $\sigma \in \mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(s)$,

$$\mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(\delta \circ \gamma)(\sigma) = \sigma \iff \bigcap_{i=1}^{k} B(\Phi_{(\delta\circ\gamma)(\tau)}(x_i), \varepsilon) \neq \varnothing, \quad \forall \tau \in [0,1],$$

$$\mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(\delta) \circ \mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(\gamma)(\sigma) = \sigma \iff \begin{cases} \bigcap_{i=1}^{k} B(\Phi_{\gamma(\tau)}(x_i), \varepsilon) \neq \varnothing, & \forall \tau \in [0,1], \\ \bigcap_{i=1}^{k} B(\Phi_{\delta(\tau)}(x_i), \varepsilon) \neq \varnothing, & \forall \tau \in [0,1]. \end{cases}$$

By definition of the composite path, these two conditions are equivalent. Therefore,

$$\mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(\delta \circ \gamma)(\sigma) = \sigma \iff \mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(\delta) \circ \mathscr{C}^{\mathrm{dyn}}_{\Phi,\varepsilon}(\gamma)(\sigma) = \sigma,$$

so the induced simplicial maps coincide, preserving composition. The case of degenerate subsimplices is analogous and immediate. □

By uniformly defining the dynamic Čech complex as a functor and clearly distinguishing between the full and sampled versions, we have established a rigorous and practical theoretical framework, laying a solid mathematical foundation for dynamic TDA.

To gain deeper insight into the evolution mechanism of topological features, we go beyond continuity and study the infinitesimal behavior of evolution. Infinitesimal generation theory provides a differential description of topological evolution, allowing us to precisely characterize topological changes in the language of differential equations.

Definition 3.11 (Tangent Space of the Complex Flow). At a point $K \in \mathcal{S}(M)$, the tangent space of the space of complexes is defined by:

$$T_K\mathcal{S}(M) := \varprojlim_{U \ni K} \mathrm{Hom}(K, K_U),$$

where K_U is a complex deformed within U, and the limit ranges over all neighborhoods of K. An element $v \in T_K\mathcal{S}(M)$ is called an infinitesimal deformation of K.

The notion of tangent space allows us to define derivatives and differentials on the space of complexes, providing tools for studying the instantaneous behavior of topological evolution. The next theorem shows that the "instantaneous topological change" (tangent vector) of a dynamic Čech complex flow is well-defined and is uniquely determined by the instantaneous velocity of the embedding map.

Theorem 3.12 (Infinitesimal Description of Dynamic Complex Flows)**.** *The dynamic Čech continuous flow $\mathscr{C}_{\Phi,\varepsilon}$ is differentiable at t, and its tangent vector satisfies*

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{C}_{\Phi,\varepsilon}(t) \in T_{\mathscr{C}_{\Phi,\varepsilon}(t)}\mathcal{S}(M).$$

This tangent vector is uniquely determined by the embedding velocity field $v_t = \dfrac{\partial \Phi}{\partial t}$, with the explicit expression:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{C}_{\Phi,\varepsilon}(t) = \mathcal{P}\left(\bigcup_{x\in M} \mathrm{d}\Phi_t(x)(v_t(x))\right),$$

where

$$\mathcal{P}\colon \mathcal{D}(\mathcal{P}) = \left\{ \bigcup_{x\in M} w_t(x) \middle| w_t(x) \in T_{\Phi_t(x)}\mathbb{R}^D \right\} \to T_{\mathscr{C}_{\Phi,\varepsilon}(t)}\mathcal{S}(M)$$

denotes the projection operator onto the tangent space.

Proof. Let $K_t = \mathscr{C}_{\Phi,\varepsilon}(t) \in \mathcal{S}(M)$. We expand the tangent space:

$$T_{K_t}\mathcal{S}(M) = \prod_{U\in\mathcal{U}_{K_t}} \mathrm{Hom}(K_t, K_U) \Big/ \sim,$$

where $\mathcal{U}_{K_t} = \{U \subset \mathcal{S}(M) \mid U \text{ is a Whitney open neighborhood of } K_t\}$, $K_U \in U$ is any complex in U, and the equivalence relation $\sim$ is

$$(f_U) \sim (g_U) \iff \exists V \in \mathcal{U}_{K_t},\ \forall U \subseteq V,\ f_U = g_U.$$

The equivalence class is denoted $[(f_U)] = \{(g_U) \in \prod \mathrm{Hom}(K_t, K_U) \mid (g_U) \sim (f_U)\}$.

Linearized velocity: $w_t(x) = \mathrm{d}\Phi_t(x)(v_t(x)) \in T_{\Phi_t(x)}\mathbb{R}^D \cong \mathbb{R}^D$. We prove that $\mathcal{P}(\bigcup w_t(x)) = [(f_U)]$ for some compatible family of homomorphisms (f_U).

First, the complex flow is differentiable at t if and only if there exists a unique $\dot{K}_t \in T_{K_t}\mathcal{S}(M)$ such that

$$\lim_{h\to 0} \frac{[(f_U^h)] - [(f_U^0)]}{h} = \dot{K}_t,$$

where $f_U^h \in \mathrm{Hom}(K_t, K_{t+h})$, $f_U^0 = \mathrm{id}_{K_t}$, and the limit is taken in the quotient topology of $T_{K_t}\mathcal{S}(M)$.

We construct the induced family (f_U^h). For any $U \in \mathcal{U}_{K_t}$, by $\Phi \in C^\infty(M \times T, \mathbb{R}^D)$, there exists $\delta_U > 0$ such that for $|h| < \delta_U$, $\Phi_{t+h}(x) = \Phi_t(x) + h \cdot w_t(x) + o(h)$ and $K_{t+h} \in U$.

For any simplex $\sigma = \{x_1, \ldots, x_k\} \in K_t$, there exists $z \in \bigcap_{i=1}^{k} B(\Phi_t(x_i), \varepsilon)$, so $d(\Phi_t(x_i), z) < \varepsilon$. Then

$$d(\Phi_{t+h}(x_i), z) \leqslant d(\Phi_t(x_i), z) + h \cdot \|w_t(x_i)\| + \|o(h)\| < \varepsilon + hC + \|o(h)\|,$$

where $C = \max_{x\in M} \|w_t(x)\| < \infty$ (since M is compact and w_t continuous). Set $\delta = \min\left\{\delta_U, \dfrac{\varepsilon - d(\Phi_t(x_i), z)}{2C}\right\}$. For $|h| < \delta$, $d(\Phi_{t+h}(x_i), z) < \varepsilon$, so $\sigma \in K_{t+h}$.

Define $f_U^h : K_t \to K_{t+h}$ to be the simplicial homomorphism with vertex map $f_U^h(x) = x$ for all $x \in K_t^{(0)}$ (where $K_t^{(0)}$ is the vertex set of K_t) and simplex extension $f_U^h(\sigma) = \sigma$ for all $\sigma \in K_t$. Clearly $f_U^h \in \mathrm{Hom}(K_t, K_U)$, so $(f_U^h)_{U\in\mathcal{U}_{K_t}} \in \prod \mathrm{Hom}(K_t, K_U)$.

We now establish the existence of the limiting homomorphism family (f_U). Define $(f_U) \in \prod \mathrm{Hom}(K_t, K_U)$ such that, for any $U \in \mathcal{U}_{K_t}$ and $\sigma \in K_t$,

$$f_U(\sigma) = \lim_{h\to 0} \frac{f_U^h(\sigma) - \sigma}{h}.$$

We clarify the geometric realizations:

1. For any k-simplex $\sigma = \{x_0, x_1, \ldots, x_k\} \in K_t$, its geometric realization is the convex set in $\mathbb{R}^D$:

$$|\sigma| = \left\{ \sum_{i=0}^{k} \lambda_i \Phi_t(x_i) \middle| \lambda_i \geqslant 0,\ \sum_{i=0}^{k} \lambda_i = 1 \right\}.$$

2. Since $f_U^h(\sigma) = \sigma$ (identity on vertices), its geometric realization is:

$$|f_U^h(\sigma)| = \left\{ \sum_{i=0}^{k} \lambda_i \Phi_{t+h}(x_i) \middle| \lambda_i \geqslant 0,\ \sum_{i=0}^{k} \lambda_i = 1 \right\}.$$

3. The geometric meaning of the difference quotient $\dfrac{f_U^h(\sigma) - \sigma}{h}$ is the "instantaneous rate of change" of the convex set:

$$\left| \frac{f_U^h(\sigma) - \sigma}{h} \right| = \left\{ \sum_{i=0}^{k} \lambda_i \cdot \frac{\Phi_{t+h}(x_i) - \Phi_t(x_i)}{h} \middle| \lambda_i \geqslant 0,\ \sum_{i=0}^{k} \lambda_i = 1 \right\}.$$

We use the Hausdorff distance to characterize convergence of convex sets. Recall that for two compact convex sets $A, B \subset \mathbb{R}^D$, the Hausdorff distance is

$$d_H(A, B) = \max \left\{ \sup_{a \in A} \inf_{b \in B} \|a - b\|,\ \sup_{b \in B} \inf_{a \in A} \|a - b\| \right\}.$$

A sequence of simplices $\{\sigma_h\}$ converges to σ if and only if their geometric realizations satisfy $d_H(|\sigma_h|, |\sigma|) \to 0$ as $h \to 0$.

First, convergence of the vertex difference quotients holds trivially. For any vertex x_i of σ, since $\Phi \in C^\infty(M \times T, \mathbb{R}^D)$, the Taylor expansion gives

$$\Phi_{t+h}(x_i) = \Phi_t(x_i) + h \cdot w_t(x_i) + o(h) \implies \exists \lim_{h \to 0} \frac{\Phi_{t+h}(x_i) - \Phi_t(x_i)}{h} = w_t(x_i).$$

This limit exists in the Euclidean topology of $\mathbb{R}^D$.

Next, convergence of the simplex difference quotients in the Hausdorff distance. We show $\left| \dfrac{f_U^h(\sigma) - \sigma}{h} \right| \xrightarrow{d_H} |f_U(\sigma)|$, where

$$|f_U(\sigma)| = \left\{ \sum_{i=0}^{k} \lambda_i w_t(x_i) \middle| \lambda_i \geqslant 0,\ \sum_{i=0}^{k} \lambda_i = 1 \right\}.$$

1. Pointwise convergence. For any $y = \sum_{i=0}^{k} \lambda_i \Phi_t(x_i) \in |\sigma|$, let $y_h = \sum_{i=0}^{k} \lambda_i \Phi_{t+h}(x_i) \in |f_U^h(\sigma)|$. Then

$$\frac{y_h - y}{h} = \sum_{i=0}^{k} \lambda_i \cdot \frac{\Phi_{t+h}(x_i) - \Phi_t(x_i)}{h} \to \sum_{i=0}^{k} \lambda_i w_t(x_i) \in |f_U(\sigma)|.$$

2. Uniform convergence. Since $|\sigma|$ is compact (convex hull of a compact set in $\mathbb{R}^D$) and $w_t : M \to \mathbb{R}^D$ is continuous, by compactness of M, w_t is uniformly continuous. Hence the vertex difference quotients $q_h(x_i) := \dfrac{\Phi_{t+h}(x_i) - \Phi_t(x_i)}{h}$ converge uniformly to $w_t(x_i)$ for $x_i \in \sigma$.

   Let

$$Q_h = \left\{ \sum_{i=0}^{k} \lambda_i q_h(x_i) \middle| \lambda_i \geqslant 0,\ \sum \lambda_i = 1 \right\},$$

$$Q = \left\{ \sum_{i=0}^{k} \lambda_i w_t(x_i) \middle| \lambda_i \geqslant 0,\ \sum \lambda_i = 1 \right\}.$$

(a) Take any $a \in Q_h$. By convexity, there exist $\lambda_i \geqslant 0$, $\sum \lambda_i = 1$ such that

$$a = \sum_{i=0}^{k} \lambda_i q_h(x_i).$$

Set $b = \sum_{i=0}^{k} \lambda_i w_t(x_i) \in Q$. By the triangle inequality and linearity of convex combinations,

$$\|a - b\| = \left\| \sum_{i=0}^{k} \lambda_i q_h(x_i) - \sum_{i=0}^{k} \lambda_i w_t(x_i) \right\| = \left\| \sum_{i=0}^{k} \lambda_i \big(q_h(x_i) - w_t(x_i)\big) \right\|.$$

By subadditivity of the norm,

$$\|a - b\| \leqslant \sum_{i=0}^{k} \lambda_i \cdot \|q_h(x_i) - w_t(x_i)\|.$$

For $|h| < \delta$, $\|q_h(x_i) - w_t(x_i)\| < \varepsilon$ for all i, and $\sum \lambda_i = 1$, so

$$\|a - b\| < \sum_{i=0}^{k} \lambda_i \cdot \varepsilon = \varepsilon.$$

Thus for every $a \in Q_h$, there exists $b \in Q$ with $\|a - b\| < \varepsilon$, so $\inf_{b \in Q} \|a - b\| < \varepsilon$. Taking the supremum yields

$$\sup_{a \in Q_h} \inf_{b \in Q} \|a - b\| \leqslant \varepsilon. \tag{3.1}$$

(b) Similarly, for every $b \in Q$, there exists $a \in Q_h$ with $\|b - a\| < \varepsilon$, so $\inf_{a \in Q_h} \|b - a\| < \varepsilon$. Taking the supremum yields

$$\sup_{b \in Q} \inf_{a \in Q_h} \|a - b\| \leqslant \varepsilon. \tag{3.2}$$

Combining (3.1), (3.2) and the definition of Hausdorff distance:

$$d_H(Q_h, Q) = \max \left\{ \sup_{a \in Q_h} \inf_{b \in Q} \|a - b\|, \sup_{b \in Q} \inf_{a \in Q_h} \|a - b\| \right\} < \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, as $h \to 0$,

$$d_H(Q_h, Q) \to 0.$$

That is, the difference quotient set Q_h converges to the target set $Q = |f_U(\sigma)|$ in the Hausdorff distance.

In summary, $f_U(\sigma) = \lim_{h \to 0} \dfrac{f_U^h(\sigma) - \sigma}{h}$ exists, with geometric realization $|f_U(\sigma)|$.

One also verifies that f_U is a simplicial homomorphism, i.e., it preserves simplex structure and inclusions. This verification is straightforward. Hence $f_U \in \mathrm{Hom}(K_t, K_U)$.

It remains to check compatibility of the homomorphism family. For any $U_2 \subset U_1$, define the restriction morphism $\rho_{U_2,U_1} : \mathrm{Hom}(K_t, K_{U_1}) \to \mathrm{Hom}(K_t, K_{U_2})$ by

$$\rho_{U_2,U_1}(f_{U_1})(\sigma) = f_{U_1}(\sigma) \cap K_{U_2}.$$

Since $U_2 \subset U_1$, the simplices of K_{U_2} are a subset of those of K_{U_1}, and $|f_{U_2}(\sigma)| = |f_{U_1}(\sigma)| \cap |K_{U_2}|$, so $f_{U_2}(\sigma) = \rho_{U_2,U_1}(f_{U_1})(\sigma)$, i.e., $f_{U_2} = \rho_{U_2,U_1}(f_{U_1})$.

Thus $(f_U)_{U \in \mathfrak{U}_{K_t}} \in \prod_{U \in \mathfrak{U}_{K_t}} \mathrm{Hom}(K_t, K_U)$ is a compatible family, as required.

We now confirm differentiability. By the equivalence relation $\sim$, $[(f_U^h)] = [(f_U^0) + h \cdot (f_U) + (o(h))]$, where $(o(h))$ denotes a family of homomorphisms each component of which is $o(h)$. By definition of convergence in the quotient topology,

$$\lim_{h \to 0} \frac{[(f_U^h)] - [(f_U^0)]}{h} = [(f_U)] \in T_{K_t}\mathcal{S}(M).$$

Hence $\mathscr{C}_{\Phi,\varepsilon}$ is differentiable at t, and we set $\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{C}_{\Phi,\varepsilon}(t) = [(f_U)]$.

Finally, we establish membership and uniqueness of the tangent vector. By construction, $[(f_U)]$ is an equivalence class in the inverse limit, so $\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{C}_{\Phi,\varepsilon}(t) \in T_{K_t}\mathcal{S}(M)$.

For uniqueness, suppose $\dot{K}_t^{(1)} = [(f_U^{(1)})]$ and $\dot{K}_t^{(2)} = [(f_U^{(2)})]$ both satisfy the definition of differentiability. Then

$$\lim_{h \to 0} \frac{[(f_U^h)] - [(f_U^0)]}{h} = [(f_U^{(1)})] = [(f_U^{(2)})].$$

By $\sim$, there exists $V \in \mathfrak{U}_{K_t}$ such that for all $U \subseteq V$, $f_U^{(1)} = f_U^{(2)}$. Since both $f_U^{(1)}$ and $f_U^{(2)}$ are induced by $w_t(x) = \mathrm{d}\Phi_t(x)(v_t(x))$, and $v_t(x) = \frac{\partial \Phi}{\partial t}(x,t)$ is unique, $w_t(x)$ is unique, hence f_U^h is unique, so $f_U^{(1)} = f_U^{(2)}$ for all $U \in \mathfrak{U}_{K_t}$, i.e., $[(f_U^{(1)})] = [(f_U^{(2)})]$. Thus the tangent vector is uniquely determined by v_t.

We now examine the projection operator $\mathcal{P}$. For any input $W = \bigcup_{x \in M} w_t(x) \in \mathcal{D}(\mathcal{P})$, define $\mathcal{P}(W) = [(f_U^W)]$, where (f_U^W) is the compatible family induced by W, such that for any $U \in \mathfrak{U}_{K_t}$,

$$f_U^W(\sigma) = \lim_{h \to 0} \frac{f_U^{h,W}(\sigma) - \sigma}{h},$$

where $f_U^{h,W}$ is the "approximate homomorphism" generated using velocities $w_t(x)$ from W (similarly to f_U^h, only replacing the velocity source with W), and $[(f_U^W)]$ is the equivalence class of (f_U^W) in the tangent space.

When the input is $W = \bigcup_{x \in M} \mathrm{d}\Phi_t(x)(v_t(x))$, the induced family (f_U^W) coincides exactly with the family (f_U) constructed in the differentiability proof, i.e., $f_U^W = f_U$. Since (f_U) is compatible, its equivalence class $[(f_U)] \in T_{K_t}\mathcal{S}(M)$, so $\mathcal{P}(W) = [(f_U)]$ is well-defined.

We have shown that the tangent vector of the complex flow is $\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{C}_{\Phi,\varepsilon}(t) = [(f_U)]$, and the projection gives $\mathcal{P}(W) = [(f_U)]$. Therefore,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{C}_{\Phi,\varepsilon}(t) = \mathcal{P}\left(\bigcup_{x \in M} \mathrm{d}\Phi_t(x)(v_t(x))\right).$$

In conclusion, the dynamic Čech continuous flow $\mathscr{C}_{\Phi,\varepsilon}$ is differentiable at t, its tangent vector lies in $T_{K_t}\mathcal{S}(M)$, is uniquely determined by the embedding velocity field v_t, and satisfies the projection formula. □

In this section, we have developed a rigorous mathematical framework for dynamic Čech complexes. Starting from the concept of continuous complex flows, via functorial constructions and tangent space theory, we have provided new tools for understanding the continuous evolution of topological structures. In subsequent sections, we build sheaf-theoretic frameworks and cohomology theories upon these foundations.

3.2 Construction of the Dynamic Čech Sheaf

In dynamic Topological Data Analysis (TDA), the traditional sheaf-theoretic framework faces fundamental challenges: how to describe the local-global relationships evolving continuously with parameters? This section will construct the theory of dynamic Čech sheaves, which provides a new mathematical language for understanding the continuous changes of topological features by endogenously encoding parametric evolution into the sheaf structure.

The core idea of constructing the dynamic Čech presheaf is to incorporate parametric dependence into the classical Čech construction, such that local sections can reflect the continuous evolution of topological structures.

Definition 3.13 (Dynamic Čech Presheaf). Let (M, g) be a d-dimensional compact Riemannian manifold, the parameter space T be a connected topological space, and $\Phi : M \times T \to \mathbb{R}^D$ be a dynamic embedding family, given a radius $\varepsilon > 0$. The dynamic Čech presheaf $\mathscr{P}_{\Phi,\varepsilon}$ is defined as follows:

For any open set $U \subseteq M \times T$,

$$\mathscr{P}_{\Phi,\varepsilon}(U) := \left\{ s : U \to \mathbb{R} \,\middle|\, \begin{array}{l} \forall (x_0, t_0) \in U, \exists \text{ open neighborhood } V \times W \subseteq U, \text{ such that } s|_{V\times W} \in C^\infty(V \times W), \\ \text{and } \forall t \in W,\ s(\cdot, t) \text{ is locally constant on each simplex of } \mathscr{C}_{\Phi,\varepsilon}(t) \end{array} \right\}$$

The restriction morphisms are the natural restrictions of functions.

This definition requires sections to maintain the local constancy determined by the Čech complex in the spatial direction and smoothness in the parametric direction, thereby capturing the continuous essence of topological evolution.

Dynamic Čech presheaves usually do not satisfy the gluing axiom of sheaves, so we need to obtain a genuine sheaf structure through sheafification.

Definition 3.14 (Dynamic Čech Sheaf). The sheafification $\mathscr{D}_{\Phi,\varepsilon} := (\mathscr{P}_{\Phi,\varepsilon})^\#$ of the dynamic Čech presheaf $\mathscr{P}_{\Phi,\varepsilon}$ forms a sheaf on $M \times T$, called the dynamic Čech sheaf. Specifically,

$$\mathscr{D}_{\Phi,\varepsilon}(U) = \left\{ s : U \to \coprod_{(x,t)\in U} (\mathscr{P}_{\Phi,\varepsilon})_{(x,t)} \,\middle|\, \begin{array}{l} \forall (x,t) \in U, \exists \text{ open neighborhood } V \ni (x,t) \text{and a section} \\ f \in \mathscr{P}_{\Phi,\varepsilon}(V), \text{ such that for all} \xi \in V,\ s(\xi) = f_\xi. \end{array} \right\}$$

The universal property of the dynamic Čech sheaf establishes its central position in the category of sheaves satisfying the dynamic local constancy condition.

Theorem 3.15 (Universal Property)**.** *Let $\mathscr{F}$ be a sheaf on $M \times T$. If for any open set $U \subseteq M \times T$ and any section $s \in \mathscr{F}(U)$, there exists an open cover $\{U_i\}$ and sections $f_i \in \mathscr{P}_{\Phi,\varepsilon}(U_i)$ such that $s|_{U_i} = f_i$ in the sheaf $\mathscr{F}$, then there exists a unique sheaf morphism $\varphi : \mathscr{F} \to \mathscr{D}_{\Phi,\varepsilon}$.*

Proof. We prove this by constructing the sheaf morphism $\varphi : \mathscr{F} \to \mathscr{D}_{\Phi,\varepsilon}$.

For any open set $U \subseteq M \times T$ and $s \in \mathscr{F}(U)$, by assumption, there exists an open cover $\{U_i\}$ and $f_i \in \mathscr{P}_{\Phi,\varepsilon}(U_i)$ such that $s|_{U_i} = f_i$ in $\mathscr{F}$. Consider the sheafification morphism $\theta : \mathscr{P}_{\Phi,\varepsilon} \to \mathscr{D}_{\Phi,\varepsilon}$, and define $\tilde{f}_i = \theta_{U_i}(f_i) \in \mathscr{D}_{\Phi,\varepsilon}(U_i)$. By the definition of sheafification, $\tilde{f}_i$ is the germ section induced by f_i, i.e., for any $(x, t) \in U_i$, $\tilde{f}_i(x, t) = (f_i)_{(x,t)}$.

On the intersection $U_i \cap U_j$, we have $s|_{U_i \cap U_j} = f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ in $\mathscr{F}$. Since $\mathscr{F}$ is a sheaf, f_i and f_j define the same section on $U_i \cap U_j$.

By the definition of the dynamic Čech presheaf, $f_i|_{U_i \cap U_j}$ and $f_j|_{U_i \cap U_j}$ are equal in $\mathscr{P}_{\Phi,\varepsilon}(U_i \cap U_j)$, so they induce the same germ section $\theta_{U_i \cap U_j}(f_i|_{U_i \cap U_j}) = \theta_{U_i \cap U_j}(f_j|_{U_i \cap U_j})$. Thus $\tilde{f}_i|_{U_i \cap U_j} = \tilde{f}_j|_{U_i \cap U_j}$. By the sheaf axioms of $\mathscr{D}_{\Phi,\varepsilon}$, there exists a unique $\tilde{s} \in \mathscr{D}_{\Phi,\varepsilon}(U)$ such that $\tilde{s}|_{U_i} = \tilde{f}_i$.

Define $\varphi_U(s) = \tilde{s}$.

It is clear that the φ defined in this way is indeed a sheaf morphism. Next, we verify the uniqueness.

Let $\psi : \mathscr{F} \to \mathscr{D}_{\Phi,\varepsilon}$ be another sheaf morphism satisfying the conditions. For any $s \in \mathscr{F}(U)$, consider the open cover $\{U_i\}$ and $f_i \in \mathscr{P}_{\Phi,\varepsilon}(U_i)$ such that $s|_{U_i} = f_i$. On U_i, $\varphi(s|_{U_i}) = \theta_{U_i}(f_i) = \psi(s|_{U_i})$. By locality, $\varphi(s) = \psi(s)$. □

Corollary 3.16. *If there exists another sheaf $\mathscr{G}$ satisfying the same universal property, then $\mathscr{G} \cong \mathscr{D}_{\Phi,\varepsilon}$.*

Proof. By the universal property of $\mathscr{G}$, there exists a unique sheaf morphism $\varphi : \mathscr{D}_{\Phi,\varepsilon} \to \mathscr{G}$. By the universal property of $\mathscr{D}_{\Phi,\varepsilon}$, there exists a unique sheaf morphism $\psi : \mathscr{G} \to \mathscr{D}_{\Phi,\varepsilon}$.

For any $s \in \mathscr{D}_{\Phi,\varepsilon}(U)$, by the local dynamic local constancy condition, there exists an open cover $\{U_i\}$ and $f_i \in \mathscr{P}_{\Phi,\varepsilon}(U_i)$ such that $s|_{U_i} = \theta_{U_i}(f_i)$. Then $(\psi \circ \varphi)(s|_{U_i}) = \psi(\varphi(\theta_{U_i}(f_i))) = \psi(\theta_{U_i}(f_i)) = \theta_{U_i}(f_i) = s|_{U_i}$, so $\psi \circ \varphi = \mathrm{id}$.

Similarly, $\varphi \circ \psi = \mathrm{id}_{\mathscr{G}}$. Therefore $\mathscr{G} \cong \mathscr{D}_{\Phi,\varepsilon}$. □

The following concept of dynamic star neighborhood generalizes the classical star neighborhood to the parametric case, providing a geometric foundation for understanding local consistency.

Definition 3.17 (Dynamic Star Neighborhood). The dynamic star neighborhood of a point $(x,t) \in M \times T$ is defined as:

$$\mathrm{St}_{\Phi,\varepsilon}(x,t) := \{(y,\tau) \in M \times T | \{x,y\} \in \mathscr{C}_{\Phi,\varepsilon}(\tau)\}$$

i.e., all pairs of points that are covered by the same ε-ball as x at parameter τ.

Proposition 3.18 (Local Consistency Condition)**.** *On the dynamic star neighborhood* $\mathrm{St}_{\Phi,\varepsilon}(x,t)$*, the sections of the dynamic Čech sheaf satisfy local consistency:*

$$s(y,\tau) = s(x,t) + O(|\tau - t| + d_M(x,y))$$

i.e., the variation of section values is jointly controlled by parametric variation and spatial distance.

Proof. Let $s \in \mathscr{D}_{\Phi,\varepsilon}(U)$ be a section of the dynamic Čech sheaf on an open set $U \subseteq M \times T$. According to the definition of the dynamic Čech sheaf, for any $(x,t) \in U$, there exists an open neighborhood $V \ni (x,t)$ and a presheaf section $f \in \mathscr{P}_{\Phi,\varepsilon}(V)$ such that for any $\xi \in V$, $s(\xi) = f_\xi$, where f_ξ is the germ of f at ξ.

By the definition of the dynamic Čech presheaf, f satisfies: for any $(y,\tau) \in V$, $f(y,\tau)$ is locally constant on each simplex of $\mathscr{C}_{\Phi,\varepsilon}(\tau)$, and f is smooth on V.

Consider $(y,\tau) \in \mathrm{St}_{\Phi,\varepsilon}(x,t) \cap V$. By the definition of the dynamic star neighborhood, $\{x,y\} \in \mathscr{C}_{\Phi,\varepsilon}(\tau)$, i.e., there exists $z \in \mathbb{R}^D$ such that:

$$z \in B(\Phi_\tau(x),\varepsilon) \cap B(\Phi_\tau(y),\varepsilon)$$

Since $f(\cdot,\tau)$ is locally constant on each simplex of $\mathscr{C}_{\Phi,\varepsilon}(\tau)$, and $\{x,y\}$ is a 1-simplex in $\mathscr{C}_{\Phi,\varepsilon}(\tau)$, we have:

$$f(x,\tau) = f(y,\tau)$$

Since $f \in C^\infty(V)$, at the point (x,t), for $(y,\tau) \in \mathrm{St}_{\Phi,\varepsilon}(x,t) \cap V$, we have:

$$f(y,\tau) = f(x,t) + \frac{\partial f}{\partial t}(x,t)(\tau - t) + \nabla_x f(x,t) \cdot (y - x) + R(y,\tau)$$

where the remainder $R(y,\tau)$ satisfies $|R(y,\tau)| \leqslant C(|\tau - t|^2 + d_M(x,y)^2)$. In particular:

$$f(y,\tau) = f(x,\tau) = f(x,t) + \frac{\partial f}{\partial t}(x,t)(\tau - t) + R(x,\tau)$$

Since f is locally constant in the spatial direction, $\nabla_x f(x,t) = 0$, so:

$$f(y,\tau) = f(x,t) + \frac{\partial f}{\partial t}(x,t)(\tau - t) + O(|\tau - t|^2)$$

Since $s(\xi) = f_\xi$ for all $\xi \in V$, in particular:

$$s(y,\tau) = f(y,\tau), \quad s(x,t) = f(x,t)$$

Therefore,

$$s(y,\tau) = s(x,t) + \frac{\partial f}{\partial t}(x,t)(\tau - t) + O(|\tau - t|^2)$$

Since $\frac{\partial f}{\partial t}(x,t)$ is bounded (as f is smooth and $M \times T$ is compact), we obtain:

$$s(y,\tau) = s(x,t) + O(|\tau - t|)$$

Considering that on the dynamic star neighborhood, $d_M(x,y)$ may not be zero, but by the Lipschitz property of the embedding and the compactness of M,

$$d_M(x,y) \leqslant L^{-1} d_{\mathbb{R}^D}(\Phi_\tau(x), \Phi_\tau(y)) < 2\varepsilon L^{-1},$$

we finally get $s(y,\tau) = s(x,t) + O(|\tau - t| + d_M(x,y))$. □

The local consistency condition quantifies the rate of change of dynamic Čech sheaf sections during topological evolution, and shows that the influence of parametric variation is linear ($O(|\tau - t|)$) and the influence of spatial distance is also linear ($O(d_M(x,y))$).

A direct corollary of the local consistency condition is the continuity of dynamic Čech sheaf sections:

Corollary 3.19 (Continuity)**.** *The sections of the dynamic Čech sheaf are continuous on the dynamic star neighborhood. In particular, if $(y_n,\tau_n) \to (x,t)$ in $\mathrm{St}_{\Phi,\varepsilon}(x,t)$, then $s(y_n,\tau_n) \to s(x,t)$.*

Proof. By the local consistency condition, when $(y_n,\tau_n) \to (x,t)$, we have $|\tau_n - t| \to 0$ and $d_M(y_n,x) \to 0$, so:

$$|s(y_n,\tau_n) - s(x,t)| \leqslant C(|\tau_n - t| + d_M(y_n,x)) \to 0$$

i.e., $s(y_n,\tau_n) \to s(x,t)$. □

The Theorem of Limit Representation of Stalks reveals the profound algebraic properties of the local structure of dynamic Čech sheaves, establishing a bridge between dynamic sheaf theory and classical smooth function theory.

Theorem 3.20 (Limit Representation of Stalks)**.** *The stalk of the dynamic Čech sheaf at (x,t) is isomorphic to:*

$$(\mathscr{D}_{\Phi,\varepsilon})_{(x,t)} \cong \varinjlim_{U \ni (x,t)} \left(C^\infty(U) \big/ \sim_{\Phi,\varepsilon} \right)$$

where the equivalence relation $\sim_{\Phi,\varepsilon}$ is defined as: $f \sim g$ if and only if there exists a neighborhood V of (x,t) such that for any $(\xi,\tau) \in V$,

$$f(\xi,\tau) = g(\xi,\tau) \quad on\, \mathrm{St}_{\Phi,\varepsilon}(\xi,\tau)$$

Proof. First, consider the left-hand side:

$$(\mathscr{D}_{\Phi,\varepsilon})_{(x,t)} = \varinjlim_{U \ni (x,t)} \mathscr{D}_{\Phi,\varepsilon}(U)$$

Restriction homomorphism: For $U_2 \subseteq U_1$, there exists $\rho_{U_2,U_1} : \mathscr{D}_{\Phi,\varepsilon}(U_1) \to \mathscr{D}_{\Phi,\varepsilon}(U_2)$, $s \mapsto s|_{U_2}$.

Equivalence relation: $s_1 \in \mathscr{D}_{\Phi,\varepsilon}(U_1) \sim s_2 \in \mathscr{D}_{\Phi,\varepsilon}(U_2)$ if and only if there exists $V \ni (x,t)$, $V \subseteq U_1 \cap U_2$, such that $s_1|_V = s_2|_V$.

Next, consider the right-hand side:

$$\varinjlim_{U \ni (x,t)} C^\infty(U) \big/ \sim_{\Phi,\varepsilon}$$

Restriction morphism: For $U_2 \subseteq U_1$, there exists $\bar{\rho}_{U_2,U_1} : C^\infty(U_1) \big/ \sim_{\Phi,\varepsilon} \to C^\infty(U_2) \big/ \sim_{\Phi,\varepsilon}$, $[f] \mapsto [f|_{U_2}]$.

Equivalence relation: $[f_1] \in C^\infty(U_1) \big/ \sim_{\Phi,\varepsilon} \sim [f_2] \in C^\infty(U_2) \big/ \sim_{\Phi,\varepsilon}$ if and only if there exists $V \ni (x,t)$, $V \subseteq U_1 \cap U_2$, such that $[f_1|_V] = [f_2|_V]$.

For each open set U containing (x,t), define the map:

$$\eta_U : \mathscr{D}_{\Phi,\varepsilon}(U) \to C^\infty(U) \big/ \sim_{\Phi,\varepsilon}, \quad s \mapsto [f_s]$$

where $f_s \in C^\infty(U)$ is the presheaf section inducing s, specifically:

By the definition of the dynamic Čech sheaf, for any $s \in \mathscr{D}_{\Phi,\varepsilon}(U)$, there exists an open cover $\{U_i\}$ and $f_i \in \mathscr{P}_{\Phi,\varepsilon}(U_i)$ such that $s|_{U_i} = \theta_{U_i}(f_i)$, where $\theta : \mathscr{P}_{\Phi,\varepsilon} \to \mathscr{D}_{\Phi,\varepsilon}$ is the sheafification morphism, and f_i satisfies $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ on the overlapping regions $U_i \cap U_j$. These f_i can be glued to form a global smooth function $f_s \in C^\infty(U)$.

η_U is well-defined:

1. If $s_1 = s_2$, then there exists a neighborhood V of (x,t) such that $s_1|_V = s_2|_V$, thus $f_{s_1} \sim_{\Phi,\varepsilon} f_{s_2}$ on V, so $[f_{s_1}] = [f_{s_2}]$.

2. The map is independent of the choice of representation: if s is induced by different f_i', then $f_s \sim_{\Phi,\varepsilon} f_s'$.

η_U is bijective:

1. Injectivity: If $\eta_U(s) = \eta_U(s')$, then $[f_s] = [f_{s'}]$, there exists $V \ni (x,t)$ such that $f_s \sim_{\Phi,\varepsilon} f_{s'}$ on V, so $s|_V = s'|_V$, thus $s_{(x,t)} = s'_{(x,t)}$.

2. Surjectivity: For any $[f] \in {C^\infty(U)}/{\sim_{\Phi,\varepsilon}}$, since $f \in \mathscr{P}_{\Phi,\varepsilon}(U)$, we obtain $s_f = \theta_U(f) \in \mathscr{D}_{\Phi,\varepsilon}(U)$ by sheafification, and $\eta_U(s_f) = [f]$.

We further claim that for any $U_2 \subseteq U_1$, the following diagram commutes:

$$\begin{tikzcd} \mathscr{D}_{\Phi,\varepsilon}(U_1) \arrow[r, "\eta_{U_1}"] \arrow[d, "\rho_{U_2,U_1}"'] & {C^\infty(U_1)}/{\sim_{\Phi,\varepsilon}} \arrow[d, "\bar{\rho}_{U_2,U_1}"] \\ \mathscr{D}_{\Phi,\varepsilon}(U_2) \arrow[r, "\eta_{U_2}"] & {C^\infty(U_2)}/{\sim_{\Phi,\varepsilon}} \end{tikzcd}$$

Specifically, for any $s \in \mathscr{D}_{\Phi,\varepsilon}(U_1)$, let $\eta_{U_1}(s) = [f_s]$, then:

$$\begin{aligned} \eta_{U_2}(\rho_{U_2,U_1}(s)) &= \eta_{U_2}(s|_{U_2}) = [f_s|_{U_2}] \\ \bar{\rho}_{U_2,U_1}(\eta_{U_1}(s)) &= \bar{\rho}_{U_2,U_1}([f_s]) = [f_s|_{U_2}] \end{aligned}$$

Thus the diagram commutes.

We have therefore established the isomorphism:

$$\eta : \{\mathscr{D}_{\Phi,\varepsilon}(U), \rho_{U',U}\} \to \left\{{C^\infty(U)}/{\sim_{\Phi,\varepsilon}}, \bar{\rho}_{U',U}\right\}$$

By the universal property of direct limits, if two systems are naturally isomorphic, their direct limits are isomorphic. Therefore:

$$(\mathscr{D}_{\Phi,\varepsilon})_{(x,t)} \cong \varinjlim_{U \ni (x,t)} {C^\infty(U)}/{\sim_{\Phi,\varepsilon}}$$

□

As a direct application of this theorem, we can obtain the dimension formula for the stalks of dynamic sheaves:

Corollary 3.21 (Dimension Formula for Stalks)**.** *At a non-critical point (x,t), the dimension of the stalk of the dynamic Čech sheaf is:*

$$\dim_{\mathbb{R}}(\mathscr{D}_{\Phi,\varepsilon})_{(x,t)} = \dim \mathrm{St}_{\Phi,\varepsilon}(x,t) + 1$$

where $\dim \mathrm{St}_{\Phi,\varepsilon}(x,t)$ *is the dimension of the dynamic star neighborhood.*

We note that the dynamic Čech sheaf is not only a sheaf, but more importantly, it is functorial, which reflects its natural compatibility with parametric evolution.

Definition 3.22 (Dynamic Sheaf Functor)*.* The construction of the dynamic Čech sheaf defines a functor from the parameter category to the sheaf category:

$$\mathscr{D}_{\Phi,\varepsilon} : T \to \mathsf{Sh}(M \times T)$$

Specifically:

1. For any $t \in T$, $\mathscr{D}_{\Phi,\varepsilon}(t) := \mathscr{D}_{\Phi,\varepsilon}|_{M\times\{t\}}$ is a sheaf on M.
2. For any path $\gamma : s \to t$, it induces a sheaf homomorphism $\mathscr{D}_{\Phi,\varepsilon}(\gamma) : \mathscr{D}_{\Phi,\varepsilon}(t) \to \mathscr{D}_{\Phi,\varepsilon}(s)$.

The dynamic Čech sheaf is well compatible with standard sheaf operations, which ensures the consistency of the theoretical framework.

Proposition 3.23 (Compatibility with Pullback Operations)**.** *Let $f : T' \to T$ be a continuous map, then there exists a natural isomorphism:*

$$f^*\mathscr{D}_{\Phi,\varepsilon} \cong \mathscr{D}_{\Phi\circ(\mathrm{id}_M\times f),\varepsilon}$$

where:

1. *f^* is the pullback sheaf, i.e., for a sheaf $\mathscr{F}$ on $M \times T$, $f^*\mathscr{F}$ is a sheaf on $M \times T'$.*
2. *$\Phi \circ (\mathrm{id}_M \times f) : M \times T' \to \mathbb{R}^D$ is the composition map, defined as $(\Phi \circ (\mathrm{id}_M \times f))(x, t') = \Phi(x, f(t'))$.*

Proof. We prove the global isomorphism by verifying the isomorphism of the two sheaves at each stalk.

Let $F = \mathrm{id}_M \times f : M \times T' \to M \times T$, and $\Phi' = \Phi \circ F$.

For any $(x, t') \in M \times T'$, consider the stalk isomorphism:

$$(f^*\mathscr{D}_{\Phi,\varepsilon})_{(x,t')} \cong (\mathscr{D}_{\Phi,\varepsilon})_{F(x,t')} = (\mathscr{D}_{\Phi,\varepsilon})_{(x,f(t'))}$$

By the Limit Representation of Stalks (Theorem 3.20):

$$(\mathscr{D}_{\Phi,\varepsilon})_{(x,f(t'))} \cong \varinjlim_{U\ni(x,f(t'))} C^\infty(U)/\sim_{\Phi,\varepsilon}$$
$$(\mathscr{D}_{\Phi',\varepsilon})_{(x,t')} \cong \varinjlim_{U'\ni(x,t')} C^\infty(U')/\sim_{\Phi',\varepsilon}$$

Consider the two direct systems:

- $\mathcal{A} = \{C^\infty(U)/\sim_{\Phi,\varepsilon}\}_{U\ni(x,f(t'))}$, where $U \subseteq M \times T$ is open.
- $\mathcal{B} = \{C^\infty(U')/\sim_{\Phi',\varepsilon}\}_{U'\ni(x,t')}$, where $U' \subseteq M \times T'$ is open.

Define the functor $\varphi : \mathcal{A} \to \mathcal{B}$ as follows:

For any $U \ni (x, f(t'))$, let $U' = F^{-1}(U)$, and define:

$$\varphi_U : C^\infty(U)/\sim_{\Phi,\varepsilon} \to C^\infty(U')/\sim_{\Phi',\varepsilon}, \quad [g] \mapsto [g \circ F]$$

It is straightforward to claim that the φ_U defined above is a well-defined linear isomorphism.

For any inclusion relation $U_1 \subseteq U_2$, consider the diagram:

$$\begin{tikzcd}
C^\infty(U_2)/\sim_{\Phi,\varepsilon} \arrow[r, "\varphi_{U_2}"] \arrow[d, "\rho_{U_1,U_2}"'] & C^\infty(F^{-1}(U_2))/\sim_{\Phi',\varepsilon} \arrow[d, "\rho_{F^{-1}(U_1),F^{-1}(U_2)}"] \\
C^\infty(U_1)/\sim_{\Phi,\varepsilon} \arrow[r, "\varphi_{U_1}"] & C^\infty(F^{-1}(U_1))/\sim_{\Phi',\varepsilon}
\end{tikzcd}$$

Commutativity follows directly from the definition of restriction morphisms.

Since φ is a natural equivalence of direct systems, by the universal property of direct limits, it induces an isomorphism of stalks:

$$(f^*\mathscr{D}_{\Phi,\varepsilon})_{(x,t')} \cong (\mathscr{D}_{\Phi,\varepsilon})_{(x,f(t'))} \cong (\mathscr{D}_{\Phi',\varepsilon})_{(x,t')}$$

Take a basis of local coordinate neighborhoods of $M \times T'$. On each coordinate neighborhood, the stalk isomorphisms are consistent on overlapping regions by stalk isomorphism, so they can be glued to form a global sheaf isomorphism.

Finally, we verify the naturality of the obtained isomorphism with respect to pullback operations. For any continuous map $g : T'' \to T'$, consider the composition $f \circ g : T'' \to T$, and we need to verify the commutativity of the diagram:

$$\begin{tikzcd}
(f \circ g)^* \mathscr{D}_{\Phi,\varepsilon} \arrow[r, "\cong"] \arrow[d, "\cong"'] & g^* f^* \mathscr{D}_{\Phi,\varepsilon} \arrow[d, "\cong"] \\
\mathscr{D}_{\Phi \circ (\mathrm{id}_M \times (f \circ g)),\varepsilon} \arrow[r, "\cong"] & g^* \mathscr{D}_{\Phi \circ (\mathrm{id}_M \times f),\varepsilon}
\end{tikzcd}$$

Commutativity is obvious from the functoriality of inductive limits and the definition of dynamic sheaves. □

This result shows that the transformation of the parameter space is compatible with the construction of dynamic sheaves.

Corollary 3.24. *Let $f : T' \to T$ be a homeomorphism, then the dynamic topological features are preserved under parametric transformation, i.e.:*

$$\mathscr{D}_{\Phi,\varepsilon} \cong f_* \mathscr{D}_{\Phi',\varepsilon}$$

where f_* is the pushforward sheaf.

Similarly, we have:

Proposition 3.25 (Compatibility with Pushforward Operations)**.** *Let $\pi : M \times T \to T$ be the projection map, then the stalks of the pushforward sheaf $\pi_* \mathscr{D}_{\Phi,\varepsilon}$ satisfy:*

$$(\pi_* \mathscr{D}_{\Phi,\varepsilon})_t \cong \Gamma(M, \mathscr{D}_{\Phi,\varepsilon}(t))$$

i.e., the stalk at each parameter t corresponds to the global sections on the fiber.

Comparisons between different dynamic Čech sheaves are realized through dynamic sheaf morphisms defined as follows.

Definition 3.26 (Dynamic Sheaf Morphism). Let $\mathscr{D}_{\Phi,\varepsilon}$ and $\mathscr{D}_{\Phi',\varepsilon'}$ be two dynamic Čech sheaves. A **dynamic sheaf morphism** $f : \mathscr{D}_{\Phi,\varepsilon} \to \mathscr{D}_{\Phi',\varepsilon'}$ is a family of sheaf homomorphisms $\{f_t : \mathscr{D}_{\Phi,\varepsilon}(t) \to \mathscr{D}_{\Phi',\varepsilon'}(t)\}_{t \in T}$ such that for any path $\gamma : s \to t$, the following diagram commutes:

$$\begin{tikzcd}
\mathscr{D}_{\Phi,\varepsilon}(t) \arrow[r, "f_t"] \arrow[d, "\mathscr{D}_{\Phi,\varepsilon}(\gamma)"'] & \mathscr{D}_{\Phi',\varepsilon'}(t) \arrow[d, "\mathscr{D}_{\Phi',\varepsilon'}(\gamma)"] \\
\mathscr{D}_{\Phi,\varepsilon}(s) \arrow[r, "f_s"] & \mathscr{D}_{\Phi',\varepsilon'}(s)
\end{tikzcd}$$

Dynamic sheaf morphisms enable us to compare the topological evolution behaviors under different dynamic processes (such as different embedding families or different scale parameters).

Through the development of this subsection, we have established a complete theoretical framework for dynamic Čech sheaves, from presheaf construction and sheafification process to the geometric description of stalk structures and functorial properties, providing a powerful mathematical tool for understanding the local-global relationships of dynamic topological processes. This framework lays a solid foundation for the persistent dynamic cohomology theory we will discuss next.

4 Persistent Dynamic Cohomology Theory

In dynamic topological data analysis, the classical single-parameter persistent homology theory faces a fundamental challenge: how to simultaneously describe the co-evolution of topological features along both the scale parameter ε and the time parameter t? This section establishes a

rigorous mathematical framework for persistent dynamic cohomology, resolving the difficulties of the two-parameter setting.

Recent research has shown that multi-parameter persistence modules are generically indecomposable [BS25], which brings challenges to our theory while also offering new insight. Unlike the classical single-parameter case, two-parameter persistence modules require more refined algebraic structures. This discovery motivates us to develop new theoretical tools to handle the complex phenomena in dynamic topological evolution.

4.1 Fundamental Theory of Persistent Dynamic Cohomology

Classical persistent homology is based on filtered complexes with a single scale parameter, while the dynamic setting requires the simultaneous interaction of scale and time. Inspired by the study of persistent homology on multi-grid posets [Día25], we establish the following categorical framework:

Definition 4.1 (Two-parameter Persistence Module). Let $\mathcal{P} = \mathbb{R}_{\geqslant 0} \times T$ be the scale-time parameter space equipped with the partial order:

$$(\varepsilon_1, t_1) \preceq (\varepsilon_2, t_2) \iff \varepsilon_1 \leqslant \varepsilon_2,\ t_1 \leqslant t_2.$$

A **two-parameter persistence module** is a functor:

$$M : (\mathcal{P}, \preceq) \to \mathsf{Vect}_K$$

such that for each pair (ε, t), it assigns a vector space $M(\varepsilon, t)$, and for any $(\varepsilon_1, t_1) \preceq (\varepsilon_2, t_2)$, it assigns a transition homomorphism $\varphi_{(\varepsilon_1,t_1)}^{(\varepsilon_2,t_2)}$, satisfying the functorial axioms.

Definition 4.2 (Persistent Dynamic Cohomology). Let a dynamic embedding family $\Phi : M \times T \to \mathbb{R}^D$ satisfy: for any $t_1 \leqslant t_2 \in T$ and any fixed scale $\varepsilon > 0$, the earlier dynamic Čech complex is a subcomplex of the later one, i.e., $\mathscr{C}_{\Phi,\varepsilon}(t_1) \hookrightarrow \mathscr{C}_{\Phi,\varepsilon}(t_2)$.

Under this condition, its **p-th persistent dynamic cohomology** is defined as the two-parameter persistence module:

$$\begin{aligned} \mathbb{PDH}^p(\Phi)(\varepsilon, t) &= H^p(\mathscr{C}_{\Phi,\varepsilon}(t); K), \\ \mathbb{PDH}^p(\Phi) &:= \left(\mathbb{PDH}^p(\Phi)(\varepsilon, t), \psi_{(\varepsilon_1,t_1)}^{(\varepsilon_2,t_2)}\right)_{(\varepsilon_1,t_1)\preceq(\varepsilon_2,t_2)} \end{aligned}$$

where the transition homomorphism $\psi_{(\varepsilon_1,t_1)}^{(\varepsilon_2,t_2)} : \mathbb{PDH}^p(\Phi)(\varepsilon_2, t_2) \to \mathbb{PDH}^p(\Phi)(\varepsilon_1, t_1)$ is the pullback cohomology homomorphism induced by the inclusion $\mathscr{C}_{\Phi,\varepsilon_1}(t_1) \hookrightarrow \mathscr{C}_{\Phi,\varepsilon_2}(t_2)$.

This inclusion condition guarantees the causal structure of the dynamic process: as time evolves, the topological structure does not collapse drastically but evolves smoothly. Technically, it ensures that the partial order $\preceq$ on the parameter space is compatible with the inclusion of complexes, so that the transition homomorphisms are well-defined.

Note that the transition maps of persistent dynamic cohomology are contravariant, in sharp contrast to the covariance of homology. This property reflects the essential nature of cohomology theory and provides a natural algebraic structure for the subsequent cup product operations.

Theorem 4.3 (Functoriality)**.** *The p-th persistent dynamic cohomology construction defines a functor from the category of dynamic embeddings to the category of two-parameter persistence modules:*

$$\mathbb{PDH}^p : \mathsf{DynEmb}(M, \mathbb{R}^D) \to \mathsf{PersMod}_2(K).$$

This functor preserves homotopy relations of dynamic embeddings: if $\Phi \simeq \Psi$, then $\mathbb{PDH}^p(\Phi) \cong \mathbb{PDH}^p(\Psi)$.

Proof. Let $f : \Phi \to \Psi$ be a morphism of dynamic embeddings. For each parameter (ε, t), f induces a simplicial map $f_{\varepsilon,t} : \mathscr{C}_{\Phi,\varepsilon}(t) \to \mathscr{C}_{\Psi,\varepsilon}(t)$, which further induces a cohomology pullback:

$$f^*_{\varepsilon,t} : H^p(\mathscr{C}_{\Psi,\varepsilon}(t); K) \to H^p(\mathscr{C}_{\Phi,\varepsilon}(t); K).$$

Define $\mathbb{PDH}^p(f)$ to be the natural transformation whose component at (ε, t) is $f^*_{\varepsilon,t}$.

It remains to verify naturality: for any $(\varepsilon_1, t_1) \preceq (\varepsilon_2, t_2)$, the following diagram commutes:

$$\begin{tikzcd}
H^p(\mathscr{C}_{\Psi,\varepsilon_2}(t_2); K) \arrow[r, "\psi_\Psi"] \arrow[d, "f^*_{\varepsilon_2,t_2}"'] & H^p(\mathscr{C}_{\Psi,\varepsilon_1}(t_1); K) \arrow[d, "f^*_{\varepsilon_1,t_1}"] \\
H^p(\mathscr{C}_{\Phi,\varepsilon_2}(t_2); K) \arrow[r, "\psi_\Phi"'] & H^p(\mathscr{C}_{\Phi,\varepsilon_1}(t_1); K)
\end{tikzcd}$$

This is guaranteed by the naturality of cohomology pullbacks.

The preservation of homotopy relations follows from a fundamental theorem in algebraic topology: simplicial homotopies induce isomorphisms in cohomology. Concretely, if $H : \Phi \simeq \Psi$ is a homotopy, then for each fixed (ε, t), H induces a chain homotopy, which yields a cohomology isomorphism. (See, e.g., [Hat02], p.201, Homotopy Invariance.) □

Persistent dynamic cohomology not only captures topological features but also preserves important algebraic structures that reflect the essential properties of dynamic processes.

Theorem 4.4 (Monotonicity)**.** *Under the defining conditions of persistent dynamic cohomology, the p-th persistent dynamic cohomology module satisfies monotonicity: for any chain of parameters $(\varepsilon_1, t_1) \preceq (\varepsilon_2, t_2) \preceq (\varepsilon_3, t_3)$, the transition homomorphisms satisfy*

$$\psi^{(\varepsilon_3,t_3)}_{(\varepsilon_1,t_1)} = \psi^{(\varepsilon_2,t_2)}_{(\varepsilon_1,t_1)} \circ \psi^{(\varepsilon_3,t_3)}_{(\varepsilon_2,t_2)},$$

and all transition homomorphisms are injective.

Proof. By the inclusion condition in the definition, for $\varepsilon_1 \leqslant \varepsilon_2$ and $t_1 \leqslant t_2$, we have $\mathscr{C}_{\Phi,\varepsilon_1}(t_1) \hookrightarrow \mathscr{C}_{\Phi,\varepsilon_2}(t_2)$. Thus for $(\varepsilon_1, t_1) \preceq (\varepsilon_2, t_2) \preceq (\varepsilon_3, t_3)$,

$$\mathscr{C}_{\Phi,\varepsilon_1}(t_1) \hookrightarrow \mathscr{C}_{\Phi,\varepsilon_2}(t_2) \hookrightarrow \mathscr{C}_{\Phi,\varepsilon_3}(t_3).$$

The transition homomorphisms are pullback homomorphisms: $\psi^{(\varepsilon_2,t_2)}_{(\varepsilon_1,t_1)} = i^*_{12}$ induced by $i_{12} : \mathscr{C}_1 \hookrightarrow \mathscr{C}_2$, $\psi^{(\varepsilon_3,t_3)}_{(\varepsilon_2,t_2)} = i^*_{23}$ induced by $i_{23} : \mathscr{C}_2 \hookrightarrow \mathscr{C}_3$, $\psi^{(\varepsilon_3,t_3)}_{(\varepsilon_1,t_1)} = i^*_{13}$ induced by $i_{13} = i_{23} \circ i_{12} : \mathscr{C}_1 \hookrightarrow \mathscr{C}_3$.

By contravariance of the cohomology functor, the pullback of a composition equals the composition of pullbacks in reverse order: $i^*_{13} = i^*_{12} \circ i^*_{23}$. Substituting gives $\psi^{(\varepsilon_3,t_3)}_{(\varepsilon_1,t_1)} = \psi^{(\varepsilon_2,t_2)}_{(\varepsilon_1,t_1)} \circ \psi^{(\varepsilon_3,t_3)}_{(\varepsilon_2,t_2)}$.

Injectivity follows from the fact that the pullback cohomology homomorphism induced by an inclusion map is injective in the same dimension. □

Monotonicity ensures that the evolution of topological features possesses a definite "causal structure", which is the theoretical foundation of dynamic topological analysis.

Theorem 4.5 (Sheaf Property)**.** *Under the defining conditions of persistent dynamic cohomology, the p-th persistent dynamic cohomology module admits an intrinsic sheaf structure: for any open cover $\{U_i\}$ of the parameter space $\mathcal{P}$, there is an exact sequence:*

$$0 \to \mathbb{PDH}^p(\Phi)(U) \to \prod_i \mathbb{PDH}^p(\Phi)(U_i) \to \prod_{i,j} \mathbb{PDH}^p(\Phi)(U_i \cap U_j),$$

where $\mathbb{PDH}^p(\Phi)(U) = \varprojlim_{(\varepsilon,t)\in U} \mathbb{PDH}^p(\Phi)(\varepsilon, t)$, and $U \subseteq \mathcal{P}$ is an open subset of the parameter space.

Proof. For any inclusion of open sets $V \subseteq U \subseteq \mathcal{P}$, define the restriction map:

$$r_V^U : \mathbb{PDH}^p(\Phi)(U) \to \mathbb{PDH}^p(\Phi)(V).$$

If $s \in \mathbb{PDH}^p(\Phi)(U)$ is a global section over U, then s is a compatible family of cohomology classes:

$$s = \{s((\varepsilon, t)) \in \mathbb{PDH}^p(\Phi)(\varepsilon, t) \mid (\varepsilon, t) \in U\},$$

such that for any $(\varepsilon_1, t_1) \preceq (\varepsilon_2, t_2) \in U$, we have $\psi_{(\varepsilon_1,t_1)}^{(\varepsilon_2,t_2)}(s((\varepsilon_2, t_2))) = s((\varepsilon_1, t_1))$. The restriction $r_V^U(s)$, written $s|_V$, is defined by $s|_V = \{s((\varepsilon, t)) \mid (\varepsilon, t) \in V\}$, which clearly satisfies compatibility over V.

By the universal property of inverse limits, the restriction maps satisfy the presheaf axioms:

1. Identity: $r_U^U(s) = s|_U = s$, i.e., $r_U^U = \mathrm{id}$;
2. Transitivity: for $W \subseteq V \subseteq U$, $r_W^U(s) = (s|_V)|_W = r_W^V \circ r_V^U(s)$, i.e., $r_W^U = r_W^V \circ r_V^U$.

Therefore, $\mathbb{PDH}^p(\Phi)$ forms a presheaf over $\mathcal{P}$ via the above restriction maps.

Define the left-hand map $\rho : \mathbb{PDH}^p(\Phi)(U) \to \prod_i \mathbb{PDH}^p(\Phi)(U_i), \quad s \mapsto (s|_{U_i})_i$, which restricts a global section s to each open set U_i in the cover, yielding a family of local sections $(s|_{U_i})_i$. We now show that ρ is injective (corresponding to the uniqueness axiom of sheaves).

Let $s, t \in \mathbb{PDH}^p(\Phi)(U)$ satisfy $\rho(s) = \rho(t)$, meaning $s|_{U_i} = t|_{U_i}$ for all i. For any $(\varepsilon, t) \in U$, since $\{U_i\}$ covers U, there exists some U_i such that $(\varepsilon, t) \in U_i$, so:

$$s((\varepsilon, t)) = (s|_{U_i})((\varepsilon, t)) = (t|_{U_i})((\varepsilon, t)) = t((\varepsilon, t)).$$

Since (ε, t) is arbitrary, $s = t$, so ρ is injective and the sequence is exact at $\mathbb{PDH}^p(\Phi)(U)$.

Define the right-hand map $\delta : \prod_i \mathbb{PDH}^p(\Phi)(U_i) \to \prod_{i,j} \mathbb{PDH}^p(\Phi)(U_i \cap U_j)$ by:

$$\delta((s_i)_i) = \left((s_i|_{U_i \cap U_j}) - (s_j|_{U_i \cap U_j})\right)_{i,j},$$

where $-$ denotes pointwise subtraction.

We prove $\mathrm{im}(\rho) = \ker(\delta)$ (corresponding to the gluing axiom of sheaves), i.e., both inclusions $\mathrm{im}(\rho) \subseteq \ker(\delta)$ and $\ker(\delta) \subseteq \mathrm{im}(\rho)$.

1. $\mathrm{im}(\rho) \subseteq \ker(\delta)$.

   For any $s \in \mathbb{PDH}^p(\Phi)(U)$, $\rho(s) = (s|_{U_i})_i$. By transitivity of restrictions, $s|_{U_i}|_{U_i \cap U_j} = s|_{U_i \cap U_j}$ and similarly $s|_{U_j}|_{U_i \cap U_j} = s|_{U_i \cap U_j}$. Thus for any $(\varepsilon, t) \in U_i \cap U_j$:

   $$(s|_{U_i}|_{U_i \cap U_j})((\varepsilon, t)) - (s|_{U_j}|_{U_i \cap U_j})((\varepsilon, t)) = s((\varepsilon, t)) - s((\varepsilon, t)) = 0,$$

   so $\delta(\rho(s)) = 0$ and hence $\mathrm{im}(\rho) \subseteq \ker(\delta)$.

2. $\ker(\delta) \subseteq \mathrm{im}(\rho)$.

   Let $(s_i)_i \in \ker(\delta)$. Then for all i, j and $(\varepsilon, t) \in U_i \cap U_j$,

   $$(s_i|_{U_i \cap U_j})((\varepsilon, t)) = (s_j|_{U_i \cap U_j})((\varepsilon, t)).$$

   Construct a global section $s \in \mathbb{PDH}^p(\Phi)(U)$: for any $(\varepsilon, t) \in U$, choose an open set U_i containing (ε, t) and set $s((\varepsilon, t)) = s_i((\varepsilon, t))$.

   Two things must be checked:

(a) Well-definedness.
If $(\varepsilon, t) \in U_i \cap U_j$, then by the definition of $\ker(\delta)$, $(s_i|_{U_i \cap U_j})((\varepsilon, t)) = (s_j|_{U_i \cap U_j})((\varepsilon, t))$. Since $s_i((\varepsilon, t)) = (s_i|_{U_i \cap U_j})((\varepsilon, t))$ and $s_j((\varepsilon, t)) = (s_j|_{U_i \cap U_j})((\varepsilon, t))$, we have $s_i((\varepsilon, t)) = s_j((\varepsilon, t))$, independent of the choice of U_i.

(b) Compatibility.
For any $(\varepsilon_1, t_1) \preceq (\varepsilon_2, t_2) \in U$, by the monotonicity condition for persistent dynamic cohomology $(\mathscr{C}_{\Phi,\varepsilon_1}(t_1) \hookrightarrow \mathscr{C}_{\Phi,\varepsilon_2}(t_2))$, the transition map $\psi^{(\varepsilon_2,t_2)}_{(\varepsilon_1,t_1)}$ is well-defined. Choose U_k containing both (ε_1, t_1) and (ε_2, t_2) (such U_k exists because U is open and the cover is exhaustive). Since s_k is a section over U_k (hence compatible), we have:

$$\psi^{(\varepsilon_2,t_2)}_{(\varepsilon_1,t_1)}(s((\varepsilon_2, t_2))) = \psi^{(\varepsilon_2,t_2)}_{(\varepsilon_1,t_1)}(s_k((\varepsilon_2, t_2))) = s_k((\varepsilon_1, t_1)) = s((\varepsilon_1, t_1)).$$

Therefore, s is a well-defined global section over U, and for any i, $s|_{U_i} = s_i$, meaning $(s_i)_i = \rho(s) \in \mathrm{im}(\rho)$. Thus $\ker(\delta) \subseteq \mathrm{im}(\rho)$.

Finally, by construction of inverse limits, $\mathbb{PDH}^p(\Phi)(U)$ satisfies:

1. For every open set U, $\mathbb{PDH}^p(\Phi)(U)$ is a K-vector space;
2. The transition homomorphisms $\psi^{(\varepsilon_2,t_2)}_{(\varepsilon_1,t_1)}$ are linear and satisfy transitivity (as shown in the monotonicity theorem);
3. The restriction maps r^U_V are compatible with transition homomorphisms, ensuring globality after gluing local sections.

In summary, the sequence

$$0 \to \mathbb{PDH}^p(\Phi)(U) \xrightarrow{\rho} \prod_i \mathbb{PDH}^p(\Phi)(U_i) \xrightarrow{\delta} \prod_{i,j} \mathbb{PDH}^p(\Phi)(U_i \cap U_j)$$

is exact, so $\mathbb{PDH}^p(\Phi)$ forms a sheaf over $\mathcal{P}$. □

The sheaf property reflects the local–global principle in dynamic topological analysis and provides a theoretical foundation for distributed computation of dynamic topological features.

4.2 Convergence, Stability, and Dynamic Topological Feature Measures

The core of classical persistent homology is barcode theory, but in the two-parameter setting we face a fundamental challenge: recent work by Bauer and Scoccola [BS25] shows that multi-parameter persistence modules are generically indecomposable, meaning the traditional barcode decomposition no longer applies.

Theorem 4.6 (Multi-parameter Indecomposability)**.** *Let Φ be a generic dynamic embedding family. Then its p-th persistent dynamic cohomology module $\mathbb{PDH}^p(\Phi)$ is* ***indecomposable****, i.e., it admits no non-trivial direct sum decomposition into interval modules.*

Proof. The proof relies on the generic indecomposability of multi-parameter persistence modules [BS25]. We verify that persistent dynamic cohomology modules satisfy the conditions of that theory:

1. $\mathbb{PDH}^p(\Phi)$ is a finitely presented multi-parameter persistence module.

   Since M is a compact manifold and the dynamic embedding Φ is continuous, for any fixed $(\varepsilon, t) \in \mathcal{P}$, the Čech complex $\mathscr{C}_{\Phi,\varepsilon}(t)$ generated by an ε-neighborhood cover is a finite simplicial complex, so its p-th cohomology group $H^p(\mathscr{C}_{\Phi,\varepsilon}(t); K)$ is a finite-dimensional K-vector space. Furthermore, by monotonicity of the dynamic embedding ($\mathscr{C}_{\Phi,\varepsilon_1}(t_1) \hookrightarrow \mathscr{C}_{\Phi,\varepsilon_2}(t_2)$ whenever $\varepsilon_1 \leqslant \varepsilon_2, t_1 \leqslant t_2$), there exists a finite scale-time grid $\mathscr{P} \subseteq \mathcal{P}$ such that $\mathbb{PDH}^p(\Phi)$ is isomorphic to the left Kan extension of its restriction to $\mathscr{P}$ (i.e., $\mathbb{PDH}^p(\Phi) \cong \mathrm{Lan}(\mathbb{PDH}^p(\Phi)|_{\mathscr{P}})$). By [BS25] Lemma 2.1 (finitely presented modules are equivalent to finite-grid restriction–extension modules), $\mathbb{PDH}^p(\Phi)$ is finitely presented. It can therefore be expressed as the cokernel of a morphism $\bigoplus_j P_j \to \bigoplus_i P_i$ of basic modules P_i, where $\mathcal{P} = \mathbb{R}_{\geqslant 0} \times T$ is a two-parameter space ($n = 2$).

2. For generic Φ, the module $\mathbb{PDH}^p(\Phi)$ is generic.

   A "generic dynamic embedding family" is defined as one in the space of C^1 dynamic embeddings that does not belong to any lower-dimensional algebraic subset (i.e., the generic case with no instantaneous collapse of topological features and no redundant parameter dependence). By [BS25] Theorem A (indecomposable modules are dense in the space of finitely presented modules) and Theorem B (ε-indecomposable modules form an open set), the module induced by such an embedding belongs to a dense open set of indecomposable modules, hence is generic.

3. No non-trivial direct sum decomposition into interval modules.

   Suppose $\mathbb{PDH}^p(\Phi) \cong A \oplus B$ with A, B non-zero interval modules. Then A, B are finitely presented interval modules. However, [BS25] Corollary 1.4 implies that decomposable modules relying on finite-dimensional indecomposable interval modules are nowhere dense, contradicting that $\mathbb{PDH}^p(\Phi)$ is generic (lies in a dense open set). Therefore $\mathbb{PDH}^p(\Phi)$ admits no non-trivial direct sum decomposition into interval modules, i.e., it is indecomposable.

□

The profound implication of this result is that we can no longer rely on traditional barcodes to represent multi-parameter topological features and must develop new descriptive methods.

Inspired by [WLY25], we give the following definition, which uses measure-theoretic tools to describe the distribution of topological features and bypasses the indecomposability problem.

Definition 4.7 (Dynamic p-dimensional Topological Feature Measure). Under the defining conditions of persistent dynamic cohomology, the dynamic topological feature measure of p-th persistent dynamic cohomology is defined as:

$$\mu_\Phi^p : \mathcal{B}(\mathcal{P}) \to \mathbb{R}_{\geqslant 0},$$

where $\mathcal{B}(\mathcal{P})$ is the Borel σ-algebra of the parameter space $\mathcal{P} = \mathbb{R}_{\geqslant 0} \times T$, and for any measurable set $A \subseteq \mathcal{P}$, the measure is given by the normalized integral:

$$\mu_\Phi^p(A) = \begin{cases} \dfrac{1}{\mathcal{R}_\Phi^p} \cdot \displaystyle\int_T \int_{\mathbb{R}_{\geqslant 0}} \dim\left(\mathbb{PDH}^p(\Phi)(\varepsilon, t)\right) \cdot \chi_A(\varepsilon, t)\, \mathrm{d}\varepsilon \mathrm{d}t, & \text{if } \mathcal{R}_\Phi^p \neq 0, \\ 0, & \text{if } \mathcal{R}_\Phi^p = 0, \end{cases}$$

where $\mathcal{R}_\Phi^p := \displaystyle\int_T \int_{\mathbb{R}_{\geqslant 0}} \dim\left(\mathbb{PDH}^p(\Phi)(\varepsilon, t)\right) \mathrm{d}\varepsilon \mathrm{d}t$ is the total integral average of the dimension of the p-th cohomology group (the normalization factor).

Due to the indecomposability of multi-parameter persistence modules, we cannot decompose $\mathbb{PDH}^p(\Phi)$ into a direct sum of interval modules as in the single-parameter case. The dynamic topological feature measure provides an alternative description, capturing the "density distribution" of topological features over the parameter space rather than discrete "lifetimes".

Stability is the key guarantee for the practical utility of persistent homology. In the two-parameter setting, we introduce a new metric tool.

Definition 4.8 (Dynamic Interpolation Distance). Let $M, N : \mathcal{P} \to \mathsf{Vec}_K$ be two two-parameter persistence modules. Their dynamic interpolation distance is defined as:

$$d_{DI}(M, N) = \inf_{\gamma} \sup_{(\varepsilon,t)\in\mathcal{P}} \|\gamma_{(\varepsilon,t)}\|_{\mathrm{op}},$$

where $\|\cdot\|_{\mathrm{op}}$ denotes the operator norm of linear maps between finite-dimensional vector spaces, and γ ranges over all contravariant natural transformations from M to N satisfying:

1. For any $(\varepsilon, t) \in \mathcal{P}$, $\gamma_{(\varepsilon,t)} : M(\varepsilon, t) \to N(\varepsilon, t)$ is a K-linear map;
2. For any pair $(\varepsilon', t') \preceq (\varepsilon, t)$, the transition maps of M and N go from (ε, t) to (ε', t') and are compatible with γ:

$$\begin{tikzcd} M(\varepsilon,t) \arrow[r, "\gamma_{(\varepsilon,t)}"] \arrow[d, "M_{(\varepsilon,t)\to(\varepsilon',t')}"'] & N(\varepsilon,t) \arrow[d, "N_{(\varepsilon,t)\to(\varepsilon',t')}"] \\ M(\varepsilon',t') \arrow[r, "\gamma_{(\varepsilon',t')}"'] & N(\varepsilon',t') \end{tikzcd}$$

   Here $M_{(\varepsilon,t)\to(\varepsilon',t')} : M(\varepsilon, t) \to M(\varepsilon', t')$ and $N_{(\varepsilon,t)\to(\varepsilon',t')} : N(\varepsilon, t) \to N(\varepsilon', t')$ are the transition morphisms of the contravariant persistence modules.

When $M = N$, the identity natural transformation $\gamma = \mathrm{id}$ satisfies $\|\gamma_{(\varepsilon,t)}\|_{\mathrm{op}} = 1$, so $d_{DI}(M, M) = 0$.

Theorem 4.9 (Stability Theorem)**.** *Under the defining conditions of persistent dynamic cohomology, the p-th persistent dynamic cohomology is Lipschitz stable with respect to the dynamic interpolation distance: for any two dynamic embedding families Φ and Ψ,*

$$d_{DI}(\mathbb{PDH}^p(\Phi), \mathbb{PDH}^p(\Psi)) \leqslant C \cdot d_{C^1}(\Phi, \Psi),$$

where

$$d_{C^1}(\Phi, \Psi) = \sup_{t\in T, x\in M} (\|\Phi(t, x) - \Psi(t, x)\|_{\mathbb{R}^D} + \|\mathbf{J}\Phi(t, x) - \mathbf{J}\Psi(t, x)\|_{\mathrm{op}})$$

is the C^1 dynamic distance, and C is a constant depending only on the manifold M and the parameter space T.

The stability theorem ensures that our theory is robust to noise and small perturbations in the data.

Theorem 4.10 (Convergence Theorem)**.** *Under the defining conditions of persistent dynamic cohomology, let $\{\Phi_n\}$ be a sequence of dynamic embedding families converging to Φ in the C^1 topology. Then the corresponding p-th persistent dynamic cohomology modules converge:*

$$\lim_{n\to\infty} \mathbb{PDH}^p(\Phi_n) = \mathbb{PDH}^p(\Phi)$$

with convergence in the dynamic interpolation distance.

The convergence theorem provides a theoretical foundation for numerical approximation, ensuring that discrete approximations preserve topological features in the limit.

Although the indecomposability of multi-parameter persistence modules limits the application of traditional barcode methods, dynamic topological feature measures and stability theory still provide powerful tools for analyzing dynamic topological features. These tools do not depend on decomposition theorems but directly on the global structural properties of persistence modules.

4.3 Statistics and Analysis of Dynamic Topological Features

Having established a rigorous mathematical framework for dynamic topological feature measures and proved their stability, we now explore the rich geometric and statistical information encoded in these measures. This section develops a measure-based framework for dynamic topological analysis, introducing moment analysis, correlation measures, concentration phenomena, and related concepts, offering a new perspective for understanding dynamic topological evolution.

Based on the dynamic topological feature measure μ_Φ^p, we can introduce various statistical moments to describe the distributional properties of topological features, providing powerful tools for quantifying topological evolution.

Definition 4.11 (Moments of Topological Features). The k-th moment of the dynamic topological feature measure μ_Φ^p is defined as:

$$M_k^p(\Phi) = \int_{\mathcal{P}} d((\varepsilon, t), (\varepsilon_0, t_0))^k \mathrm{d}\mu_\Phi^p(\varepsilon, t),$$

where d is a suitable metric on $\mathcal{P}$ (e.g., Euclidean), and (ε_0, t_0) is a reference point (e.g., the origin or the barycenter of the distribution).

In fact, it is clear that:

- The first moment M_1^p describes the "center of mass" of the topological features;
- The second moment M_2^p describes their dispersion.

Specifically, we may define directional moments by

$$M_{1,\varepsilon}^p := \int_{\mathcal{P}} d((\varepsilon, t_0), (\varepsilon_0, t_0)) \mathrm{d}\mu_\Phi^p(\varepsilon, t), \quad M_{1,t}^p := \int_{\mathcal{P}} d((\varepsilon_0, t), (\varepsilon_0, t_0)) \mathrm{d}\mu_\Phi^p(\varepsilon, t).$$

The first moments then have clear geometric meaning:

1. A larger $M_{1,\varepsilon}^p$ indicates that p-dimensional topological features appear mainly at large scales;
2. $M_{1,t}^p$ reflects the average time of appearance along the time axis.

Similarly, the magnitude of M_2^p quantifies the concentration of topological features over the parameter space.

To compare topological features between different dynamic processes, we introduce the following Wasserstein distance to measure the discrepancy between p-dimensional topological feature distributions. For background on Wasserstein distances, we refer to the survey by Victor M. Panaretos and Yoav Zemel (2019) [PZ19], and in particular, the following definition is inspired by Eq. (2) in Section 1.2 with $p = 1$.

Definition 4.12 (1st-order Wasserstein Topological Distance). Let dynamic embedding families $\Phi, \Psi : T \times M \to \mathbb{R}^D$ have p-dimensional dynamic topological feature measures $\mu^p_\Phi, \mu^p_\Psi : \mathcal{B}(\mathcal{P}) \to \mathbb{R}_{\geqslant 0}$, both being probability measures[2], i.e., $\mu^p_\Phi(\mathcal{P}) = \mu^p_\Psi(\mathcal{P}) = 1$.

The 1st-order Wasserstein topological distance between Φ and Ψ is defined by

$$d_W^{1,p}(\Phi, \Psi) = \inf_{\pi \in \Pi(\mu^p_\Phi, \mu^p_\Psi)} \int_{\mathcal{P}\times\mathcal{P}} d_{\mathcal{P}}\left((\varepsilon_1, t_1), (\varepsilon_2, t_2)\right) \mathrm{d}\pi(\varepsilon_1, t_1; \varepsilon_2, t_2),$$

where the metric $d_{\mathcal{P}}$ on the parameter space is the weighted product Euclidean metric[3]

$$d_{\mathcal{P}}\left((\varepsilon_1, t_1), (\varepsilon_2, t_2)\right) = \sqrt{\omega_\varepsilon \cdot (\varepsilon_1 - \varepsilon_2)^2 + \omega_t \cdot d_T(t_1, t_2)^2},$$

with $\omega_\varepsilon > 0$, $\omega_t > 0$ weight coefficients, and where $d_T(t_1, t_2) = |t_1 - t_2|$ if $T \subseteq \mathbb{R}$.

The set of couplings $\Pi(\mu^p_\Phi, \mu^p_\Psi)$ consists of all joint probability measures with marginals μ^p_Φ and μ^p_Ψ, i.e., for all $\pi \in \Pi(\mu^p_\Phi, \mu^p_\Psi)$,

$$\int_{\mathcal{P}} \mathrm{d}\pi(\varepsilon_1, t_1; \varepsilon_2, t_2) = \mu^p_\Phi(\mathrm{d}\varepsilon_1, \mathrm{d}t_1), \quad \int_{\mathcal{P}} \mathrm{d}\pi(\varepsilon_1, t_1; \varepsilon_2, t_2) = \mu^p_\Psi(\mathrm{d}\varepsilon_2, \mathrm{d}t_2).$$

Theorem 4.13 (Stability of the Wasserstein Topological Distance)**.** *Let $\Phi, \Psi : T \times M \to \mathbb{R}^D$ be C^1 dynamic embedding families, and let μ^p_Φ, μ^p_Ψ be their p-dimensional dynamic topological feature probability measures. Then the 1st-order Wasserstein topological distance is Lipschitz stable with respect to the C^1 dynamic distance d_{C^1}:*

$$d_W^{1,p}(\Phi, \Psi) \leqslant C \cdot d_{C^1}(\Phi, \Psi),$$

where $C > 0$ is a constant depending only on M, T, D, and the weights $\omega_\varepsilon, \omega_t$ in the parameter space metric $d_{\mathcal{P}}$, independent of Φ and Ψ.

In dynamic topological data analysis, topological features at different scales often encode distinct geometric and topological information. Fine-scale (small ε) features reflect local geometric structure, while coarse-scale (large ε) features reveal global topological properties of the data. To systematically characterize the distribution of topological features across scales, we introduce the following scale spectrum.

Definition 4.14 (Scale Spectrum of Topological Features). For a dynamic embedding family Φ, the scale spectrum of its p-dimensional topological features is the function $S^p_\Phi : \mathbb{R}_{\geqslant 0} \to [0, 1]$ given by

$$S^p_\Phi(\varepsilon_0) = \mu^p_\Phi\left(\{(\varepsilon, t) \in \mathcal{P} \mid \varepsilon \leqslant \varepsilon_0\}\right),$$

where μ^p_Φ is the dynamic topological feature measure (a Borel probability measure).

Since $\{(\varepsilon, t) \in \mathcal{P} \mid \varepsilon \leqslant \varepsilon_0\} \in \mathcal{B}(\mathcal{P})$, the measure is well-defined.

The scale spectrum $S^p_\Phi(\varepsilon_0)$ has an intuitive meaning: it is the total proportion of p-dimensional topological features with scale at most ε_0 over the full parameter space $\mathcal{P}$.

Since $S^p_\Phi(\varepsilon_0)$ is essentially the marginal cumulative distribution function of topological features over the scale parameter ε (ignoring variation in time), it increases monotonically from 0 to 1 as ε_0 goes from 0 to $+\infty$, giving clear geometric intuition:

1. If $S^p_\Phi(\varepsilon_0)$ rapidly approaches 1 at small ε_0, then p-dimensional topological features are concentrated at fine scales;

[2] This satisfies the prerequisite for Wasserstein distances, see [PZ19], Section 1.2.

[3] Adapted to the product structure of $\mathcal{P}$, consistent with the general metric space formulation in the survey, see [PZ19], Section 1.2.

2. If $S^p_\Phi(\varepsilon_0)$ approaches 1 slowly at large ε_0, then topological features remain significant at coarse scales.

This visually encodes the scale preference of dynamic topological features.

Proposition 4.15 (Properties of the Scale Spectrum)**.** *The scale spectrum $S^p_\Phi(\varepsilon_0)$ is a non-decreasing function of ε_0, and*

$$\lim_{\varepsilon_0 \to 0} S^p_\Phi(\varepsilon_0) = 0, \quad \lim_{\varepsilon_0 \to \infty} S^p_\Phi(\varepsilon_0) = 1.$$

5 Dynamic Topological Feature Extraction Algorithms

Based on the persistent dynamic cohomology theoretical framework established in the previous sections, this section presents a set of algorithms to transform abstract mathematical theories into computable practical tools. In particular, addressing the indecomposability of multi-parameter persistent modules, we develop measure-based topological feature extraction and visualization methods.

5.1 Efficient Construction Algorithm for Dynamic Čech Complexes

In dynamic topological data analysis, the construction of dynamic Čech complexes is the foundation for computing persistent dynamic cohomology. According to Definition 2.4 and Definition 3.1, we need to construct the corresponding Čech complex $\mathscr{C}_{\Phi,\varepsilon}(t)$ for each time point $t \in T$. Traditional Čech complex construction suffers from the combinatorial explosion problem with a time complexity of $O(n^{d+1})$ (see [EH21] for detailed explanations), where n is the number of points and d is the manifold dimension. This problem is particularly severe in the dynamic setting.

Remark 5.1. Based on Theorem 3.5, the continuity of the dynamic embedding family Φ ensures that the Čech complexes at adjacent time points change minimally, making the incremental update strategy effective.

Using KD-trees or ball trees for neighborhood queries reduces the neighborhood search complexity from $O(n)$ to $O(\log n)$.

Proposition 5.2 (Complexity Analysis)**.** *Let the total size of the dynamic point cloud be $n = |X_t|$ (assuming approximately stable size across time steps), the dynamic change ratio $\delta \in (0, 1]$ be defined as the ratio of the total number of changed points to the total number of points, i.e., $\delta n = |\Delta X^{add}| + |\Delta X^{remove}| + |\Delta X^{move}|$, the manifold dimension be d, and the maximum simplex dimension be $d_{max} \leqslant d$. Then the amortized time complexity of algorithm 1 is:*

$$O\left((\delta n)^{d_{max}+1} + n \log n\right)$$

Proof. $O(n \log n)$ is a one-time cost in the initial phase (basic cost of static complex construction and KD-tree/ball tree index construction).

$O((\delta n)^{d_{\max}+1})$ is the dominant cost for each incremental update (deletion of affected simplices and generation/validation of new candidate simplices). □

Compared with the time complexity $O(n^{d_{\max}+1})$ of "reconstructing the static Čech complex from scratch at each time step", when the dynamic change ratio $\delta \ll 1$ (small change scenarios), the complexity of this algorithm decreases exponentially, achieving significant efficiency improvement.

algorithm 1 provides an efficient geometric foundation for subsequent persistent dynamic cohomology computations, ensuring the feasibility of the entire dynamic topological analysis pipeline.

Algorithm 1: Incremental Construction Algorithm for Dynamic Čech Complexes

Input: Dynamic point cloud sequence $\{X_t\}_{t=1}^{T} \subset M$; `// M is a compact smooth manifold`
Scale parameter $\varepsilon > 0$; `// Radius parameter for Čech complex`
Maximum dimension $d_{\max} \in \mathbb{N}$; `// Highest dimension of simplices to construct`
Movement threshold $\delta > 0$; `// Distance threshold for determining point movement`

Output: Dynamic Čech complex sequence $\{\mathscr{C}_{\Phi,\varepsilon}(t)\}_{t=1}^{T}$; `// Φ is the dynamic embedding family`

Initialization:

Construct KD-tree (or ball tree) index $\mathcal{I}_1$ for initial point cloud X_1; `// Complexity` $O(n \log n)$

$\mathscr{C}_{\Phi,\varepsilon}(1) \leftarrow \text{ComputeStaticCech}(X_1, \varepsilon, d_{\max})$; `// All valid simplices in Čech complex at` $t = 1$

$\mathcal{I} \leftarrow \mathcal{I}_1$; `// Initialize index cache`

for $t \leftarrow 2$ **to** T **do**

`/* Detect point cloud changes (based on dynamic embedding Φ) */`

$\Delta X^{\text{add}} \leftarrow \{x \in X_t \mid x \notin X_{t-1}\}$; `// Set of newly added points`

$\Delta X^{\text{remove}} \leftarrow \{x \in X_{t-1} \mid x \notin X_t\}$; `// Set of removed points`

$\Delta X^{\text{move}} \leftarrow \{x \in X_t \cap X_{t-1} \mid \|\Phi(t,x) - \Phi(t-1,x)\|_{\mathbb{R}^N} > \delta\}$; `// Set of moved points`

`/* Incrementally update spatial index */`

Remove index entries corresponding to ΔX^{remove} from $\mathcal{I}$;

Add index entries for ΔX^{add} to $\mathcal{I}$ (based on coordinates of $\Phi(t,x)$);

Update coordinates of ΔX^{move} in $\mathcal{I}$ (synchronize dynamic embedding positions);

$\mathscr{C}_{\Phi,\varepsilon}(t) \leftarrow \mathscr{C}_{\Phi,\varepsilon}(t-1)$; `// Inherit complex from previous time step`

`/* Simplices containing removed points are necessarily invalid (vertices do not exist) */`

Remove all simplices in $\mathscr{C}_{\Phi,\varepsilon}(t)$ that contain vertices from ΔX^{remove};

`/* Simplices affected by moved points: those containing moved points or overlapping with moved points' neighborhoods */`

AffectedSimplices $\leftarrow \{\sigma \in \mathscr{C}_{\Phi,\varepsilon}(t) \mid \sigma \cap \Delta X^{\text{move}} \neq \varnothing \text{ or } \exists x \in \Delta X^{\text{move}}, \sigma \subset N_{2\varepsilon}(x)\}$;

Remove all simplices in AffectedSimplices from $\mathscr{C}_{\Phi,\varepsilon}(t)$;

`/* Construct newly valid simplices */`

$\Delta X^{\text{affected}} \leftarrow \Delta X^{\text{add}} \cup \Delta X^{\text{move}}$; `// All affected point sets`

foreach $x \in \Delta X^{affected}$ **do**

`/* Query ε-neighborhood: based on dynamic embedding coordinates Φ(t, x) */`

$N_\varepsilon(x) \leftarrow \text{QueryNeighborhood}(\mathcal{I}, \Phi(t,x), \varepsilon)$; `// Index query, complexity` $O(\log n + k)$

$N_\varepsilon(x) \leftarrow N_\varepsilon(x) \cap X_t$; `// Filter invalid vertices (retain only current time step points)`

`/* Generate all possible simplices containing` x `(dimension` $\leqslant d_{\max}$`) */`

CandidateSimplices $\leftarrow \{\sigma \subseteq N_\varepsilon(x) \mid x \in \sigma, 1 \leqslant |\sigma| \leqslant d_{\max} + 1\}$;

foreach *candidate simplex* $\sigma \in$ *CandidateSimplices* **do**

`/* Verify Čech complex condition, i.e., intersection of ε-balls of all vertices is non-empty */`

if $\bigcap_{y \in \sigma} B(\Phi(t,y), \varepsilon) \neq \varnothing$ **then**

if $\sigma \notin \mathscr{C}_{\Phi,\varepsilon}(t)$ **then**

Add σ to $\mathscr{C}_{\Phi,\varepsilon}(t)$; `// Avoid duplicate additions`

Output $\mathscr{C}_{\Phi,\varepsilon}(t)$;

Table 1: Performance Comparison of Dynamic Čech Construction Algorithms

Method	Time Complexity	Space Complexity	Applicable Scenarios
Direct Construction	$O(n^{d+1})$	$O(n^{d+1})$	Static small-scale data
Incremental Construction	$O((\delta n)^{d+1} + n \log n)$	$O(n^d)$	Dynamic medium-to-large-scale data

5.2 Multi-Parameter Computation Method for Persistent Dynamic Cohomology

Traditional computation of single-parameter persistent homology relies on the interval decomposition theorem, obtaining barcodes through boundary matrix reduction. However, as shown in Theorem 4.6, two-parameter persistent modules are generally indecomposable, rendering traditional matrix reduction methods inapplicable. We aim to develop a multi-parameter computation method independent of the decomposition theorem, based on the definition of dynamic topological feature measures.

The core computational goal is to approximate the dynamic topological feature measure μ^p_Φ. Recalling the definition, this requires computing the following integral:

$$\mathcal{R}^p_\Phi = \int_T \int_{\mathbb{R}_{\geqslant 0}} \dim \left(\mathbb{PDH}^p(\Phi)(\varepsilon, t)\right) \mathrm{d}\varepsilon \mathrm{d}t$$

as well as weighted integrals over arbitrary Borel sets $A \subseteq \mathcal{P}$.

Since $\mathbb{PDH}^p(\Phi)(\varepsilon, t) = H^p(\mathscr{C}_{\Phi,\varepsilon}(t); K)$ cannot be directly computed on the continuous parameter space, we adopt the following computational approximation strategy:

1. Discretize the continuous two-parameter space $\mathcal{P} = \mathbb{R}_{\geqslant 0} \times T$ into a finite grid as a numerical approximation to the continuous theory.
2. Compute the dimension of the cohomology group $\beta^p(\varepsilon_i, t_j) = \dim H^p(\mathscr{C}_{\Phi,\varepsilon_i}(t_j); K)$ at each grid point (ε_i, t_j), constructing discrete samples of persistent dynamic cohomology.
3. Approximate $\mathcal{R}^p_\Phi$ and $\mu^p_\Phi(A)$ via numerical integration methods to obtain numerical estimates of the topological feature measure.

Remark 5.3. We provide some possible computational optimization methods here:

1. Utilize the incremental nature of algorithm 1—when computing complexes at adjacent time points t_j and t_{j+1} for the same scale ε_i, partial computation results can be reused.
2. For a fixed time t_j, Čech complexes at different scales ε_i form a filtration, allowing standard persistent homology libraries (e.g., Phat library [A+17] or Dionysus library [Mor10]) to compute Betti numbers for all scales at once, avoiding repeated complex construction.

Proposition 5.4 (Computational Complexity)**.** *Let the parameter grid size be $N = N_\varepsilon \times N_t$, the point cloud size be n, the manifold dimension be d, and the maximum simplex dimension be d_{max}. Then the time and space complexity of algorithm 2 are:*

$$\textit{Time: } O\left(N \cdot [(\delta n)^{d_{max}+1} + n \log n + C_{homology}]\right)$$
$$\textit{Space: } O(N + n^{d_{max}})$$

Algorithm 2: Multi-Parameter Computation of Persistent Dynamic Cohomology

Input: Dynamic point cloud sequence $\{X_t\}_{t=1}^{T}$
Scale parameter range $[\varepsilon_{\min}, \varepsilon_{\max}]$, number of discrete points N_ε
Time parameter range $[t_{\min}, t_{\max}]$, number of discrete points N_t
Homology dimension p
Coefficient field K
Output: Discrete Betti number matrix $B^p \in \mathbb{R}^{N_\varepsilon \times N_t}$
Discrete approximation of dynamic topological feature measure $\hat{\mu}_\Phi^p$

```
/* Initialization                                                    */
```
$\varepsilon_{\text{grid}} \leftarrow \text{Linspace}(\varepsilon_{\min}, \varepsilon_{\max}, N_\varepsilon)$; `// Scale discretization`
$t_{\text{grid}} \leftarrow \text{Linspace}(t_{\min}, t_{\max}, N_t)$; `// Time discretization`
$B^p \leftarrow \text{ZerosMatrix}(N_\varepsilon, N_t)$; `// Initialize Betti number matrix`
for $i \leftarrow 1$ **to** N_ε **do**
  $\varepsilon \leftarrow \varepsilon_{\text{grid}}[i]$
  for $j \leftarrow 1$ **to** N_t **do**
    $t \leftarrow t_{\text{grid}}[j]$
    `/* Compute dynamic Čech complex at current parameters using algorithm 1 */`
    $\mathscr{C} \leftarrow \text{DynamicCechComplex}(X_t, \Phi, \varepsilon, t)$
    `/* Compute dimension of` p`-dimensional cohomology group of simplicial complex */`
    $B^p[i,j] \leftarrow \dim H^p(\mathscr{C}; K)$; `// Call standard homology computation library`

```
/* Compute normalization factor (numerical integration)              */
```
$\Delta\varepsilon \leftarrow (\varepsilon_{\max} - \varepsilon_{\min})/(N_\varepsilon - 1)$
$\Delta t \leftarrow (t_{\max} - t_{\min})/(N_t - 1)$
$\mathcal{R}_\Phi^p \leftarrow \sum_{i=1}^{N_\varepsilon} \sum_{j=1}^{N_t} B^p[i,j] \cdot \Delta\varepsilon \cdot \Delta t$; `// Rectangular rule numerical integration`

```
/* Define discrete measure approximation                             */
```
$$\hat{\mu}_\Phi^p(A) \leftarrow \begin{cases} \dfrac{1}{\mathcal{R}_\Phi^p} \cdot \displaystyle\sum_{(\varepsilon_i, t_j) \in A} B^p[i,j] \cdot \Delta\varepsilon \cdot \Delta t, & \text{if } \mathcal{R}_\Phi^p \neq 0 \\ 0, & \text{if } \mathcal{R}_\Phi^p = 0 \end{cases}$$
return $B^p, \hat{\mu}_\Phi^p$

where $C_{homology}$ is the complexity of computing the homology group of a single complex, which depends on the adopted algorithm and coefficient field: traditional boundary matrix reduction algorithms (e.g., Smith normal form) have a complexity of $O(m^3)$ over the integer coefficient field, while optimized algorithms over the binary field ($\mathbb{Z}_2$) can reduce the complexity to $O(m^2 \log m)$ or lower. Here $m = O(n^{d_{max}})$ denotes the total number of simplices in the complex.

Table 2: Comparison of Multi-Parameter and Single-Parameter Persistent Homology Computation Methods

Feature	Single-Parameter Persistent Homology	Multi-Parameter Persistent Dynamic Cohomology
Mathematical Foundation	Interval Decomposition Theorem	Dynamic Topological Feature Measure
Output Form	Barcodes/Persistence Diagrams	Betti Number Matrix/Measure Approximation
Algebraic Structure	Decomposable into direct sum of interval modules	Generally indecomposable
Computational Complexity	$O(m^3)$, m = complex size	$O(N \cdot (\text{complex computation} + m^3))$
Feature Description	Discrete "lifespan"	Continuous "density distribution"
Stability	Classical Stability Theorem	Theorem 4.9

Proposition 5.5 (Measure Approximation Convergence)**.** *Let $\{\mathcal{P}_N\}$ be a sequence of increasingly refined partitions of the parameter space, satisfying that the grid diameter tends to 0 as $N \to \infty$. Then the discrete measure approximation $\hat{\mu}^p_\Phi$ output by algorithm 2 converges weakly to the true dynamic topological feature measure μ^p_Φ:*

$$\hat{\mu}^p_\Phi \xrightarrow{w} \mu^p_\Phi \quad (N \to \infty)$$

Proof. The correctness of the proposition is evident, guaranteed by:

1. The Betti number function $(\varepsilon, t) \mapsto \dim H^p(\mathscr{C}_{\Phi,\varepsilon}(t); K)$ is piecewise constant on $\mathcal{P}$ and bounded at critical values where the complex changes.

2. Numerical integration converges to the true integral as the grid is refined.

3. By Lebesgue integration theory, the rectangular rule numerical integration converges for piecewise constant integrands.

□

Remark 5.6. The indecomposability of multi-parameter persistent modules manifests computationally in that we cannot obtain "two-parameter barcodes" through simple matrix operations. This paper circumvents this issue by computing Betti number distributions—while losing some algebraic information, it retains statistical information of topological features, which is sufficient for practical dynamic topological analysis applications.

This algorithm provides a feasible computational implementation for the persistent dynamic cohomology theory, enabling the mathematical framework established in section 4 to be applied to practical data analysis.

5.3 Measure-Based Topological Feature Extraction and Visualization Algorithms

Based on the dynamic topological feature measure $\hat{\mu}^p_\Phi$ and Betti number matrix B^p computed by algorithm 2, we develop a complete set of topological feature extraction and visualization methods.

Algorithm 3: Dynamic Topological Feature Extraction (Basic Feature Computation)

Input: Betti number matrix $B^p \in \mathbb{R}^{N_\varepsilon \times N_t}$
Scale grid $\varepsilon_{\text{grid}} \in \mathbb{R}^{N_\varepsilon}$
Time grid $t_{\text{grid}} \in \mathbb{R}^{N_t}$
Reference point (ε_0, t_0) ; `// Typically the center of the parameter space`
Output: Topological feature moments M^p_1, M^p_2
Scale spectrum S^p_Φ
`/* Initialization:` `*/`
total_mass $\leftarrow \sum_{i,j} B^p[i,j]$
if *total_mass* $= 0$ **then**
  return $0, 0, \text{ZerosVector}(N_\varepsilon)$; `// No topological features detected`
`/* Compute topological feature moments:` `*/`
$M^p_1, M^p_2 \leftarrow 0, 0$
foreach $(\varepsilon_i, t_j) \in$ *grid points* **do**
  weight $\leftarrow B^p[i,j]/\text{total_mass}$
  $r \leftarrow \text{Distance}((\varepsilon_i, t_j), (\varepsilon_0, t_0))$; `// Euclidean distance`
  $M^p_1 \leftarrow M^p_1 + \text{weight} \cdot r$
  $M^p_2 \leftarrow M^p_2 + \text{weight} \cdot r^2$
`/* Compute scale spectrum:` `*/`
$S^p_\Phi \leftarrow \text{ZerosVector}(N_\varepsilon)$
cumulative $\leftarrow 0$
for $i \leftarrow 1$ **to** N_ε **do**
  cumulative $\leftarrow$ cumulative $+ \sum_j B^p[i,j]$
  $S^p_\Phi[i] \leftarrow$ cumulative/total_mass
return M^p_1, M^p_2, S^p_Φ

Remark 5.7. Algorithm Implementation Notes:

- The ground distance matrix ComputeCostMatrix precomputes weighted Euclidean distances between all pairs of grid points to avoid repeated computations;
- Probability measure conversion Normalize normalizes the Betti number matrix to a probability distribution $\mu = B/\sum B$;
- Sinkhorn algorithm SinkhornWasserstein, see [Cut13] for details.

Remark 5.8. Notes on Topological Feature Visualization:

1. Betti number heatmap: Visualize the B^p matrix as a heatmap on the scale-time plane, with color intensity representing Betti number magnitude.
2. Topological feature distribution plot: Plot the probability density distribution on the parameter space $\mathcal{P}$ based on $\hat{\mu}^p_\Phi$.

Algorithm 4: Wasserstein Distance Computation (Multi-Dataset Comparison)

```
Input: Set of Betti number matrices {B^{p,(1)}, ..., B^{p,(m)}}
Scale grid ε_grid, time grid t_grid
Weight parameters (ω_ε, ω_t) ;                    // Default values: (1,1)
Output: Wasserstein distance matrix D_W ∈ R^{m×m}
/* Preprocessing:                                                    */
C ← ComputeCostMatrix(ε_grid, t_grid, ω_ε, ω_t) ;   // Precompute ground distance
 matrix
/* Compute distance matrix:                                          */
D_W ← ZerosMatrix(m, m)
for u ← 1 to m do
    μ_u ← Normalize(B^{p,(u)}) ;          // Normalize to probability measure
    for v ← u + 1 to m do
        μ_v ← Normalize(B^{p,(v)})
        D_W[u, v] ← SinkhornWasserstein(μ_u, μ_v, C)
        D_W[v, u] ← D_W[u, v] ;                                       // Symmetry
return D_W
```

3. Scale spectrum curve: Plot the curve of $S^p_\Phi(\varepsilon)$ against ε, showing the scale distribution characteristics of topological features.

4. Dynamic evolution plot: For a fixed scale ε, plot the curve of Betti numbers against time, showing the dynamic evolution of topological features.

6 Conclusion and Outlook

This paper systematically constructs a unified mathematical framework for dynamic Čech cohomology and sheaf theory on manifolds. The main contributions are as follows:

1. A rigorous mathematical definition of the dynamic Čech complex is proposed, and the concept of continuous complex flow is established, lifting dynamic topological processes from discrete approximations to the theoretical level of continuous evolution.

2. A theory of dynamic Čech sheaves is constructed. Through dynamic star neighborhoods and local consistency conditions, a local–global correspondence for the continuous evolution of topological features is established, providing a new mathematical language for dynamic topological analysis.

3. The persistent dynamic cohomology module $\mathbb{PDH}^p(\Phi)$ is defined, and dynamic topological feature measures are introduced to characterize the distribution of topological features in the scale–time parameter space, effectively avoiding the indecomposability issue of multi-parameter persistent modules.

4. Based on the above theoretical foundations, a complete algorithmic implementation is developed, including efficient construction of dynamic Čech complexes, multi-parameter computation of persistent dynamic cohomology, and measure-based topological feature extraction and visualization.

The theoretical significance of this work is mainly reflected in:

Algorithm 5: Topological Feature Visualization Generation

```
Input: Betti number matrix B^p
Scale grid ε_grid
Time grid t_grid
Visualization type V_type ∈ {heatmap, distribution, scale_spectrum, evolution}
Output: Visualization figure
/* Initialization:                                                    */
total_mass ← Σ_{i,j} B^p[i, j]
if total_mass = 0 then
    fig ← CreateEmptyPlot()
    fig.title ← "No Topological Features Detected"
    return fig
switch V_type do
    case heatmap do
        /* Generate Betti number heatmap                              */
        fig ← CreateHeatmap(B^p, ε_grid, t_grid)
        Set color map: cool colors for low Betti numbers, warm colors for high Betti
          numbers
        Add axis labels: "Scale Parameter ε", "Time Parameter t"
        Add color bar label: "dim H^p"
    case distribution do
        /* Generate topological feature probability distribution plot */
        density ← B^p/total_mass ;                    // Safe normalization
        fig ← CreateContourPlot(density, ε_grid, t_grid)
        Plot contour lines to show concentrated regions of topological features
    case scale_spectrum do
        /* Generate scale spectrum                                    */
        S ← ZerosVector(N_ε)
        cumulative ← 0
        for i ← 1 to N_ε do
            cumulative ← cumulative + Σ_j B^p[i, j]
            S[i] ← cumulative/total_mass ;                // Safe normalization
        fig ← CreateLinePlot(ε_grid, S)
        Set axis labels: "Scale ε", "Cumulative Distribution S^p_Φ(ε)"
    case evolution do
        /* Generate dynamic evolution plot                            */
        /* Automatically select representative scale:  scale with maximum
           total Betti numbers                                        */
        scale_sums ← Σ_j B^p[:, j] ;          // Total Betti numbers for each scale
        k ← argmax(scale_sums) ;              // Select most representative scale
        ε_ref ← ε_grid[k]
        evolution_curve ← B^p[k, :]
        fig ← CreateLinePlot(t_grid, evolution_curve)
        Set axis labels: "Time t", "dim H^p(ε_ref, t)"
        Set title: "Topological Evolution at Fixed Scale ε = ε_ref"
return fig
```

- Extending the classical Čech cohomology theory from the static setting to the dynamic setting, enriching the application scope of algebraic topology in data analysis.
- Building a bridge between sheaf theory and dynamic topological analysis, offering a new perspective for understanding the evolutionary mechanism of topological features.
- The proposed dynamic topological feature measure provides a new approach for the study of multi-parameter persistent homology, overcoming the theoretical obstacles caused by indecomposability.

Based on the framework established in this paper, further research can be carried out in the following directions:

1. Further taking the dimension p as an additional parameter, regarding the persistent dynamic cohomology module as a three-parameter persistent module $\mathbb{PDH}^{\bullet}(\Phi)$. The correlation between different dimensions remains a challenging problem.
2. Developing more efficient parallel computing algorithms to handle larger-scale dynamic data, such as video sequences and sensor network data.
3. Conducting applied research in biomedicine, fluid mechanics, cosmology, and other fields to explore the value of dynamic topological features in the analysis of complex systems.
4. Combining dynamic topological features with deep learning to develop topology-aware neural network models and enhance the understanding of dynamic data.
5. Investigating online dynamic topological analysis algorithms to realize real-time topological monitoring and anomaly detection for streaming data.

The unified framework of dynamic Čech cohomology and sheaf theory on manifolds established in this paper not only provides new mathematical tools for topological data analysis, but also opens up a new research direction for understanding the evolution of topological structures in dynamic systems. With the improvement of computing power and the expansion of application demands, this framework is expected to play an important role in more fields and promote the in-depth development of topological data analysis theory and applications.

A Proof of Stability Theorem 4.9 and 4.13

There exists a uniform Lipschitz constant $c > 0$ such that $\|J\Phi_t(x)\| \leqslant c$ and $\|J\Psi_t(x)\| \leqslant c$ hold for all $x \in M$ and $t \in T$.

Let $\delta = d_{C^1}(\Phi, \Psi)$. We will prove that there exists a constant $C > 0$ such that

$$d_{DI}(\mathbb{PDH}^p(\Phi), \mathbb{PDH}^p(\Psi)) \leqslant C \cdot \delta.$$

We first establish uniform boundedness of the Hausdorff distance.

For any fixed $t \in T$, consider the Hausdorff distance between the embedded images $\Phi_t(M) \subseteq \mathbb{R}^D$ and $\Psi_t(M) \subseteq \mathbb{R}^D$:

$$d_H(\Phi_t(M), \Psi_t(M)) = \max\left\{ \sup_{x\in M} \inf_{y\in M} \|\Phi_t(x) - \Psi_t(y)\|,\ \sup_{y\in M} \inf_{x\in M} \|\Phi_t(x) - \Psi_t(y)\| \right\}.$$

By C^1-continuity of the dynamic embeddings, $\|\Phi_t(x) - \Psi_t(x)\| \leqslant \delta$ for all $x \in M$. By Lipschitz continuity of Φ_t, for any $y \in M$:

$$\|\Phi_t(x) - \Psi_t(y)\| \leqslant \|\Phi_t(x) - \Phi_t(y)\| + \|\Phi_t(y) - \Psi_t(y)\| \leqslant c \cdot d_M(x, y) + \delta.$$

Taking $y = x$ yields $\inf_{y \in M} \|\Phi_t(x) - \Psi_t(y)\| \leqslant \delta$. Similarly, $\inf_{x \in M} \|\Phi_t(x) - \Psi_t(y)\| \leqslant \delta$. Therefore:

$$d_H(\Phi_t(M), \Psi_t(M)) \leqslant \delta. \tag{A.1}$$

Now, for any $(\varepsilon, t) \in \mathcal{P}$, we have the property:

$$|\mathscr{C}_{\Phi,\varepsilon}(t)| \subseteq |\mathscr{C}_{\Psi,\varepsilon+\delta}(t)|, \quad |\mathscr{C}_{\Psi,\varepsilon}(t)| \subseteq |\mathscr{C}_{\Phi,\varepsilon+\delta}(t)|,$$

where $|\cdot|$ denotes the geometric realization. For instance, the first inclusion can also be expressed as:

$$\forall \sigma \in \mathscr{C}_{\Phi,\varepsilon}(t),\ \exists \tau \in \mathscr{C}_{\Psi,\varepsilon+\delta}(t) \quad \text{s.t.} \quad \dim \sigma = \dim \tau \quad \text{and} \quad \sigma \simeq \tau.$$

A brief justification follows. Take any simplex $\sigma \in \mathscr{C}_{\Phi,\varepsilon}(t)$. By definition of the dynamic Čech complex, there exists $z \in \mathbb{R}^D$ such that $z \in \bigcap_{x \in \sigma} B(\Phi_t(x), \varepsilon)$. For any $x \in \sigma$, by the Hausdorff bound equation (A.1), there exists $y_x \in M$ such that $\|\Phi_t(x) - \Psi_t(y_x)\| \leqslant \delta$. By the triangle inequality:

$$\|\Psi_t(y_x) - z\| \leqslant \|\Psi_t(y_x) - \Phi_t(x)\| + \|\Phi_t(x) - z\| < \delta + \varepsilon.$$

Thus $z \in B(\Psi_t(y_x), \varepsilon + \delta)$. Let $\tau = \{y_x \mid x \in \sigma\}$. Then

$$z \in \bigcap_{y_x \in \tau} B(\Psi_t(y_x), \varepsilon + \delta) \neq \emptyset,$$

so $\tau \in \mathscr{C}_{\Psi,\varepsilon+\delta}(t)$. Moreover, $\dim \tau = \dim \sigma$, and by the Nerve Lemma2.22, the geometric realizations of σ and τ are homotopy equivalent $\sigma \simeq \tau$. Hence the topological structure of σ is fully preserved in $\mathscr{C}_{\Psi,\varepsilon+\delta}(t)$.

The second inclusion follows similarly.

Accordingly, there exists a simplicial morphism $f_{\Phi\to\Psi} : \mathscr{C}_{\Phi,\varepsilon}(t) \to \mathscr{C}_{\Psi,\varepsilon+\delta}(t)$ satisfying $f_{\Phi\to\Psi}(x) = y_x$ and $f_{\Phi\to\Psi}(\sigma) = \tau$.

By basic properties of algebraic topology, a simplicial morphism induces a cohomology homomorphism:

$$f^*_{\Phi\to\Psi} : H^p(\mathscr{C}_{\Phi,\varepsilon}(t); K) \to H^p(\mathscr{C}_{\Psi,\varepsilon+\delta}(t); K)\,.$$

Since $\sigma \simeq \tau$, the geometric realizations of $\mathscr{C}_{\Phi,\varepsilon}(t)$ and $\mathscr{C}_{\Psi,\varepsilon+\delta}(t)$ are homotopy equivalent, so $f^*_{\Phi\to\Psi}$ is an isomorphism.

On the other hand, by definition of persistent dynamic cohomology, the inclusion $j_\Psi : \mathscr{C}_{\Psi,\varepsilon}(t) \hookrightarrow \mathscr{C}_{\Psi,\varepsilon+\delta}(t)$ induces a transition homomorphism:

$$\psi^{(\varepsilon+\delta,t)}_{(\varepsilon,t)} : H^p(\mathscr{C}_{\Psi,\varepsilon}(t); K) \to H^p(\mathscr{C}_{\Psi,\varepsilon+\delta}(t); K)\,.$$

By monotonicity of persistent dynamic cohomology, $\psi^{(\varepsilon+\delta,t)}_{(\varepsilon,t)}$ is injective, hence its right inverse $\left(\psi^{(\varepsilon+\delta,t)}_{(\varepsilon,t)}\right)^{-1}$ exists, and its operator norm is bounded.

Define the linear map $\gamma_{(\varepsilon,t)} : \mathbb{PDH}^p(\Phi)(\varepsilon,t) \to \mathbb{PDH}^p(\Psi)(\varepsilon,t)$ by:

$$\gamma_{(\varepsilon,t)} = \left(\psi_{(\varepsilon,t)}^{(\varepsilon+\delta,t)}\right)^{-1} \circ f_{\Phi\to\Psi}^*.$$

Since M is a compact Riemannian manifold, both $\mathscr{C}_{\Phi,\varepsilon}(t)$ and $\mathscr{C}_{\Psi,\varepsilon}(t)$ are finite simplicial complexes, and their cohomology groups are finite-dimensional vector spaces. Combining the following two facts:

1. The operator norm of the isomorphism $\|f_{\Phi\to\Psi}^*\|_{\mathrm{op}} \leqslant C_1$, where C_1 is a constant depending only on the compactness of M, as the family of simplicial morphisms is uniformly bounded over the compact manifold;
2. The operator norm of the right inverse of the transition homomorphism $\left\|\left(\psi_{(\varepsilon,t)}^{(\varepsilon+\delta,t)}\right)^{-1}\right\|_{\mathrm{op}} \leqslant C_2$, where C_2 is controlled by the combinatorial complexity of $\mathscr{C}_{\Psi,\varepsilon}(t)$, hence by the dimension d and compactness of M,

by submultiplicativity of the operator norm:

$$\|\gamma_{(\varepsilon,t)}\|_{\mathrm{op}} \leqslant \left\|\left(\psi_{(\varepsilon,t)}^{(\varepsilon+\delta,t)}\right)^{-1}\right\|_{\mathrm{op}} \cdot \|f_{\Phi\to\Psi}^*\|_{\mathrm{op}} \leqslant C_1C_2. \tag{A.2}$$

We verify that $\gamma = \{\gamma_{(\varepsilon,t)}\}_{(\varepsilon,t)\in\mathcal{P}}$ satisfies functorial compatibility for the dynamic interpolation distance, i.e., for any $(\varepsilon_1,t_1) \preceq (\varepsilon_2,t_2)$, the following diagram commutes:

$$\begin{array}{ccc}
\mathbb{PDH}^p(\Phi)(\varepsilon_2,t_2) & \xrightarrow{\gamma_{(\varepsilon_2,t_2)}} & \mathbb{PDH}^p(\Psi)(\varepsilon_2,t_2) \\
\varphi_{(\varepsilon_1,t_1)}^{(\varepsilon_2,t_2)}\Big\downarrow & & \Big\downarrow\psi_{(\varepsilon_1,t_1)}^{(\varepsilon_2,t_2)} \\
\mathbb{PDH}^p(\Phi)(\varepsilon_1,t_1) & \xrightarrow[\gamma_{(\varepsilon_1,t_1)}]{} & \mathbb{PDH}^p(\Psi)(\varepsilon_1,t_1)
\end{array}$$

where the transition maps satisfy:

$$\varphi_{(\varepsilon_1,t_1)}^{(\varepsilon_2,t_2)} : \mathbb{PDH}^p(\Phi)(\varepsilon_1,t_1) \to \mathbb{PDH}^p(\Phi)(\varepsilon_2,t_2), \qquad \psi_{(\varepsilon_1,t_1)}^{(\varepsilon_2,t_2)} : \mathbb{PDH}^p(\Psi)(\varepsilon_1,t_1) \to \mathbb{PDH}^p(\Psi)(\varepsilon_2,t_2).$$

Substitute $\gamma_{(\varepsilon,t)} = \left(\psi_{(\varepsilon,t)}^{(\varepsilon+\delta,t)}\right)^{-1} \circ f_{\Phi\to\Psi}^*$ and expand both compositions:

$$\begin{aligned}
\text{Left-hand side}: \quad & \gamma_{(\varepsilon_2,t_2)} \circ \varphi_{(\varepsilon_1,t_1)}^{(\varepsilon_2,t_2)} = \left(\psi_{(\varepsilon_2,t_2)}^{(\varepsilon_2+\delta,t_2)}\right)^{-1} \circ f_{\Phi\to\Psi}^* \circ \varphi_{(\varepsilon_1,t_1)}^{(\varepsilon_2,t_2)}, \\
\text{Right-hand side}: \quad & \psi_{(\varepsilon_1,t_1)}^{(\varepsilon_2,t_2)} \circ \gamma_{(\varepsilon_1,t_1)} = \psi_{(\varepsilon_1,t_1)}^{(\varepsilon_2,t_2)} \circ \left(\psi_{(\varepsilon_1,t_1)}^{(\varepsilon_1+\delta,t_1)}\right)^{-1} \circ f_{\Phi\to\Psi}^*.
\end{aligned}$$

By naturality of transition maps in persistent dynamic cohomology—meaning that $f_{\Phi\to\Psi}$ commutes with transition maps—we have

$$f_{\Phi\to\Psi}^* \circ \varphi_{(\varepsilon_1,t_1)}^{(\varepsilon_2,t_2)} = \psi_{(\varepsilon_1+\delta,t_1)}^{(\varepsilon_2+\delta,t_2)} \circ f_{\Phi\to\Psi}^*,$$

where $\psi_{(\varepsilon_1+\delta,t_1)}^{(\varepsilon_2+\delta,t_2)} : \mathbb{PDH}^p(\Psi)(\varepsilon_1+\delta,t_1) \to \mathbb{PDH}^p(\Psi)(\varepsilon_2+\delta,t_2)$. Substituting into the left-hand side:

$$\mathrm{LHS} = \left(\psi_{(\varepsilon_2,t_2)}^{(\varepsilon_2+\delta,t_2)}\right)^{-1} \circ \psi_{(\varepsilon_1+\delta,t_1)}^{(\varepsilon_2+\delta,t_2)} \circ f_{\Phi\to\Psi}^*.$$

On the other hand, by bifunctoriality of persistent dynamic cohomology, the following diagram commutes:

$$\begin{tikzcd}
\mathbb{PDH}^p(\Psi)(\varepsilon_1, t_1) \arrow[r, "\psi^{(\varepsilon_1+\delta,t_1)}_{(\varepsilon_1,t_1)}"] \arrow[d, "\psi^{(\varepsilon_2,t_2)}_{(\varepsilon_1,t_1)}"'] & \mathbb{PDH}^p(\Psi)(\varepsilon_1+\delta, t_1) \arrow[d, "\psi^{(\varepsilon_2+\delta,t_2)}_{(\varepsilon_1+\delta,t_1)}"] \\
\mathbb{PDH}^p(\Psi)(\varepsilon_2, t_2) \arrow[r, "\psi^{(\varepsilon_2+\delta,t_2)}_{(\varepsilon_2,t_2)}"] & \mathbb{PDH}^p(\Psi)(\varepsilon_2+\delta, t_2)
\end{tikzcd}$$

Therefore LHS = RHS, and functorial compatibility is proved.

From the above construction, $\gamma = \{\gamma_{(\varepsilon,t)}\}$ is a valid family of morphisms satisfying functorial compatibility, with operator norm bounded by $\|\gamma_{(\varepsilon,t)}\|_{\mathrm{op}} \leqslant C_1 C_2$.

We now relate this norm bound to the C^1-distance $\delta = d_{C^1}(\Phi, \Psi)$.

Since M is a compact Riemannian manifold and the dynamic embedding Ψ is uniformly Lipschitz with constant c, the cohomological perturbation induced by scaling from ε to $\varepsilon + \delta$ is linearly controllable. Specifically, the operator norm of the inverse transition map satisfies

$$\left\| \left(\psi^{(\varepsilon+\delta,t)}_{(\varepsilon,t)} \right)^{-1} \right\|_{\mathrm{op}} \leqslant c' \cdot \delta,$$

where $c' > 0$ is a universal constant depending only on M and the Lipschitz constant c, independent of Φ and Ψ.

Note that C_1 is also a constant depending only on the compactness of M. Set $C = C_1 c'$, which is still a constant depending only on the ambient manifold M. Then

$$C_1 C_2 \leqslant C_1 \cdot c' \cdot \delta = C \cdot \delta.$$

Thus the operator norm of the morphism family γ satisfies the uniform bound

$$\sup_{(\varepsilon,t)\in\mathcal{P}} \|\gamma_{(\varepsilon,t)}\|_{\mathrm{op}} \leqslant C \cdot \delta = C \cdot d_{C^1}(\Phi, \Psi).$$

By definition of the dynamic interpolation distance:

$$d_{DI}(\mathbb{PDH}^p(\Phi), \mathbb{PDH}^p(\Psi)) = \inf_{\gamma} \sup_{(\varepsilon,t)\in\mathcal{P}} \|\gamma_{(\varepsilon,t)}\|_{\mathrm{op}},$$

where the infimum is taken over all functorially compatible morphism families γ. Since we have shown that γ is an admissible morphism family, its supremum gives an upper bound for the dynamic interpolation distance, so

$$d_{DI}(\mathbb{PDH}^p(\Phi), \mathbb{PDH}^p(\Psi)) \leqslant C \cdot d_{C^1}(\Phi, \Psi).$$

The stability for the first-order Wasserstein distance $d_W^{1,p}(\Phi, \Psi)$ between the dynamic topological feature measures μ_Φ^p and μ_Ψ^p follows by analogous arguments.

In summary, there exist constants $C, C' > 0$ depending only on the manifold M and the Lipschitz constant c of the dynamic embeddings such that:

$$d_{DI}\left(\mathbb{PDH}^p(\Phi), \mathbb{PDH}^p(\Psi)\right) \leqslant C \cdot d_{C^1}(\Phi, \Psi), \quad d_W^{1,p}(\Phi, \Psi) \leqslant C' \cdot d_{C^1}(\Phi, \Psi).$$

B Proof of Convergence Theorem 4.10 and its Corollaries

In the main text, the convergence theorem 4.10 is stated as follows: if a sequence of dynamic embedding families $\{\Phi_n\}$ converges to Φ in the C^1 topology, then the corresponding p-th persistent dynamic cohomology modules $\mathbb{PDH}^p(\Phi_n)$ converge to $\mathbb{PDH}^p(\Phi)$ with respect to the dynamic interpolation distance. It is natural to conjecture that the associated topological feature measures, Wasserstein distances, topological moments, and scale spectra also converge accordingly.

B.1 Convergence in the Dynamic Interpolation Distance

Recall the definition of C^1-convergence: A sequence of dynamic embedding families $\{\Phi_n\}$ converges to Φ in the C^1 topology if

$$\lim_{n\to\infty} d_{C^1}(\Phi_n, \Phi) = 0,$$

where

$$d_{C^1}(\Phi, \Psi) = \sup_{t\in T,\, x\in M} \Big(\|\Phi(t,x) - \Psi(t,x)\|_{\mathbb{R}^N} + \|\mathbf{J}\Phi(t,x) - \mathbf{J}\Psi(t,x)\|_{\mathrm{op}} \Big).$$

For any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geqslant N$, $d_{C^1}(\Phi_n, \Phi) < \varepsilon$.

By the stability theorem 4.9, there exists a universal constant $C > 0$ depending only on the manifold M, the time interval T, and the Lipschitz constant, such that for any two dynamic embeddings Ψ_1, Ψ_2,

$$d_{DI}\big(\mathbb{PDH}^p(\Psi_1), \mathbb{PDH}^p(\Psi_2)\big) \leqslant C \cdot d_{C^1}(\Psi_1, \Psi_2).$$

Substituting $\Psi_1 = \Phi_n$ and $\Psi_2 = \Phi$, we obtain

$$d_{DI}\big(\mathbb{PDH}^p(\Phi_n), \mathbb{PDH}^p(\Phi)\big) \leqslant C \cdot d_{C^1}(\Phi_n, \Phi).$$

By the definition of C^1-convergence, $\lim_{n\to\infty} d_{C^1}(\Phi_n, \Phi) = 0$. Combining the above inequality and the squeeze theorem:

$$0 \leqslant \lim_{n\to\infty} d_{DI}\big(\mathbb{PDH}^p(\Phi_n), \mathbb{PDH}^p(\Phi)\big) \leqslant C \cdot \lim_{n\to\infty} d_{C^1}(\Phi_n, \Phi) = 0.$$

Therefore,

$$\lim_{n\to\infty} d_{DI}\big(\mathbb{PDH}^p(\Phi_n), \mathbb{PDH}^p(\Phi)\big) = 0,$$

i.e., $\mathbb{PDH}^p(\Phi_n)$ converges to $\mathbb{PDH}^p(\Phi)$ in the dynamic interpolation distance.

B.2 Other Convergence Results

Convergence in the dynamic interpolation distance implies pointwise convergence of the Betti number function:

$$\lim_{n\to\infty} \beta^p_{\Phi_n}(\varepsilon, t) = \beta^p_{\Phi}(\varepsilon, t), \quad \forall (\varepsilon, t) \in \mathcal{P}.$$

Proof. By assumption, $\lim_{n\to\infty} d_{DI}\big(\mathbb{PDH}^p(\Phi_n), \mathbb{PDH}^p(\Phi)\big) = 0$, meaning that for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n > N$,

$$\sup_{(\varepsilon,t)\in\mathcal{P}} \|\gamma_{n,(\varepsilon,t)}\|_{\mathrm{op}} < \epsilon.$$

For any fixed $(\varepsilon, t) \in \mathcal{P}$, the Betti number $\beta^p_{\Phi}(\varepsilon, t) = \dim\big(\mathbb{PDH}^p(\Phi)(\varepsilon, t)\big)$ is a nonnegative integer characterizing the topological information of the cohomology group.

When n is sufficiently large, the difference in topological structure between $\mathbb{PDH}^p(\Phi_n)(\varepsilon, t)$ and $\mathbb{PDH}^p(\Phi)(\varepsilon, t)$ tends to zero, so their core topological invariant (the Betti number) must converge to the same value [4], i.e., $\lim_{n\to\infty} \beta^p_{\Phi_n}(\varepsilon, t) = \beta^p_{\Phi}(\varepsilon, t)$. □

[4] Since M is a compact manifold, the dimension of $\mathbb{PDH}^p(\Phi)(\varepsilon, t)$ is bounded (denote the upper bound by B), i.e., $0 \leqslant \beta^p_{\Phi}(\varepsilon, t) \leqslant B$. The Betti number is a discrete nonnegative integer. If the sequence $\{\beta^p_{\Phi_n}(\varepsilon, t)\}$ satisfies that the "topological difference tends to zero", then the sequence must eventually converge to the unique integer $\beta^p_{\Phi}(\varepsilon, t)$.

Since both M and T are compact manifolds, $\mathcal{P} = \mathbb{R}_{\geqslant 0} \times T$ is a locally compact, separable metric space. Moreover, there exists a constant $B > 0$ depending only on the topology of M such that for any dynamic embedding Ψ,

$$0 \leqslant \beta^p_\Psi(\varepsilon, t) \leqslant B, \quad \forall (\varepsilon, t) \in \mathcal{P}.$$

From the pointwise convergence $\beta^p_{\Phi_n} \to \beta^p_\Phi$ and Lebesgue's dominated convergence theorem, for any Borel set $A \subseteq \mathcal{P}$,

$$\lim_{n\to\infty} \int_A \beta^p_{\Phi_n}(\varepsilon, t)\mathrm{d}\varepsilon\mathrm{d}t = \int_A \beta^p_\Phi(\varepsilon, t)\mathrm{d}\varepsilon\mathrm{d}t.$$

For the normalization factor $\mathcal{R}^p_\Phi = \int_{\mathcal{P}} \beta^p_\Phi(\varepsilon, t)\mathrm{d}\varepsilon\mathrm{d}t$, taking $A = \mathcal{P}$ gives $\lim_{n\to\infty} \mathcal{R}^p_{\Phi_n} = \mathcal{R}^p_\Phi$.

Recall that the dynamic topological feature measure is defined by

$$\mu^p_{\Phi_n}(A) = \frac{1}{\mathcal{R}^p_{\Phi_n}} \int_A \beta^p_{\Phi_n}(\varepsilon, t)\mathrm{d}\varepsilon\mathrm{d}t, \quad \mu^p_\Phi(A) = \frac{1}{\mathcal{R}^p_\Phi} \int_A \beta^p_\Phi(\varepsilon, t)\mathrm{d}\varepsilon\mathrm{d}t.$$

Combining $\mathcal{R}^p_{\Phi_n} \to \mathcal{R}^p_\Phi$ and the convergence of integrals, we obtain

$$\lim_{n\to\infty} \mu^p_{\Phi_n}(A) = \frac{1}{\mathcal{R}^p_\Phi} \int_A \beta^p_\Phi(\varepsilon, t)\mathrm{d}\varepsilon\mathrm{d}t = \mu^p_\Phi(A), \quad \forall \text{ Borel } A \subseteq \mathcal{P},$$

i.e., $\mu^p_{\Phi_n}$ converges weakly to μ^p_Φ.

Since $\mathcal{P}$ is a locally compact, separable metric space, all dynamic topological feature measures $\mu^p_{\Phi_n}$ and μ^p_Φ are supported on a common compact subset of $\mathcal{P}$, and their density functions (the Betti number functions) are uniformly bounded. Together with the weak convergence of measures

$$\lim_{n\to\infty} \mu^p_{\Phi_n}(A) = \mu^p_\Phi(A), \quad \forall \text{ Borel set } A \subseteq \mathcal{P},$$

we obtain convergence of the first-order Wasserstein distance:

$$\lim_{n\to\infty} d^{1,p}_W(\Phi_n, \Phi) = 0.$$

Furthermore, for any $k \geqslant 1$, the topological moment is $M^p_k(\Phi) = \int_{\mathcal{P}} \varepsilon^k \mathrm{d}\mu^p_\Phi(\varepsilon, t)$. By weak convergence of measures and continuity of integration (ε^k is continuous and bounded),

$$\lim_{n\to\infty} M^p_k(\Phi_n) = \int_{\mathcal{P}} \varepsilon^k \mathrm{d}\mu^p_\Phi(\varepsilon, t) = M^p_k(\Phi).$$

Similarly, the scale spectrum is $S^p_\Phi(\varepsilon) = \int_T \beta^p_\Phi(\varepsilon, t)\mathrm{d}t$. For a fixed ε, by the dominated convergence theorem,

$$\lim_{n\to\infty} S^p_{\Phi_n}(\varepsilon) = \int_T \beta^p_\Phi(\varepsilon, t)\mathrm{d}t = S^p_\Phi(\varepsilon).$$

These results collectively form the convergence foundation of the persistent dynamic cohomology theory, providing solid theoretical support for numerical computation and practical applications.

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